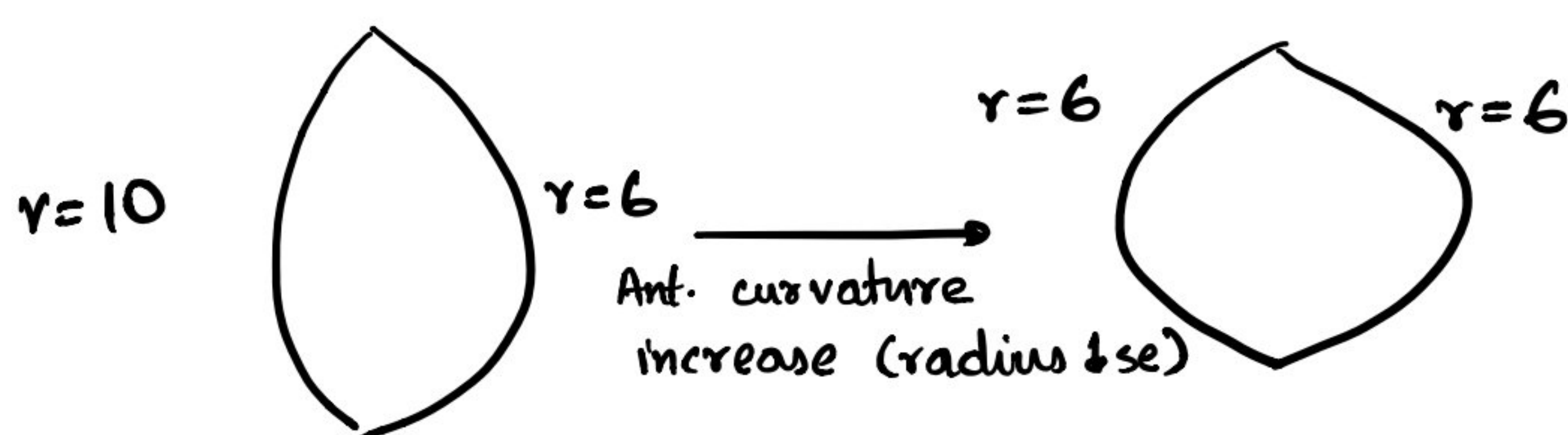
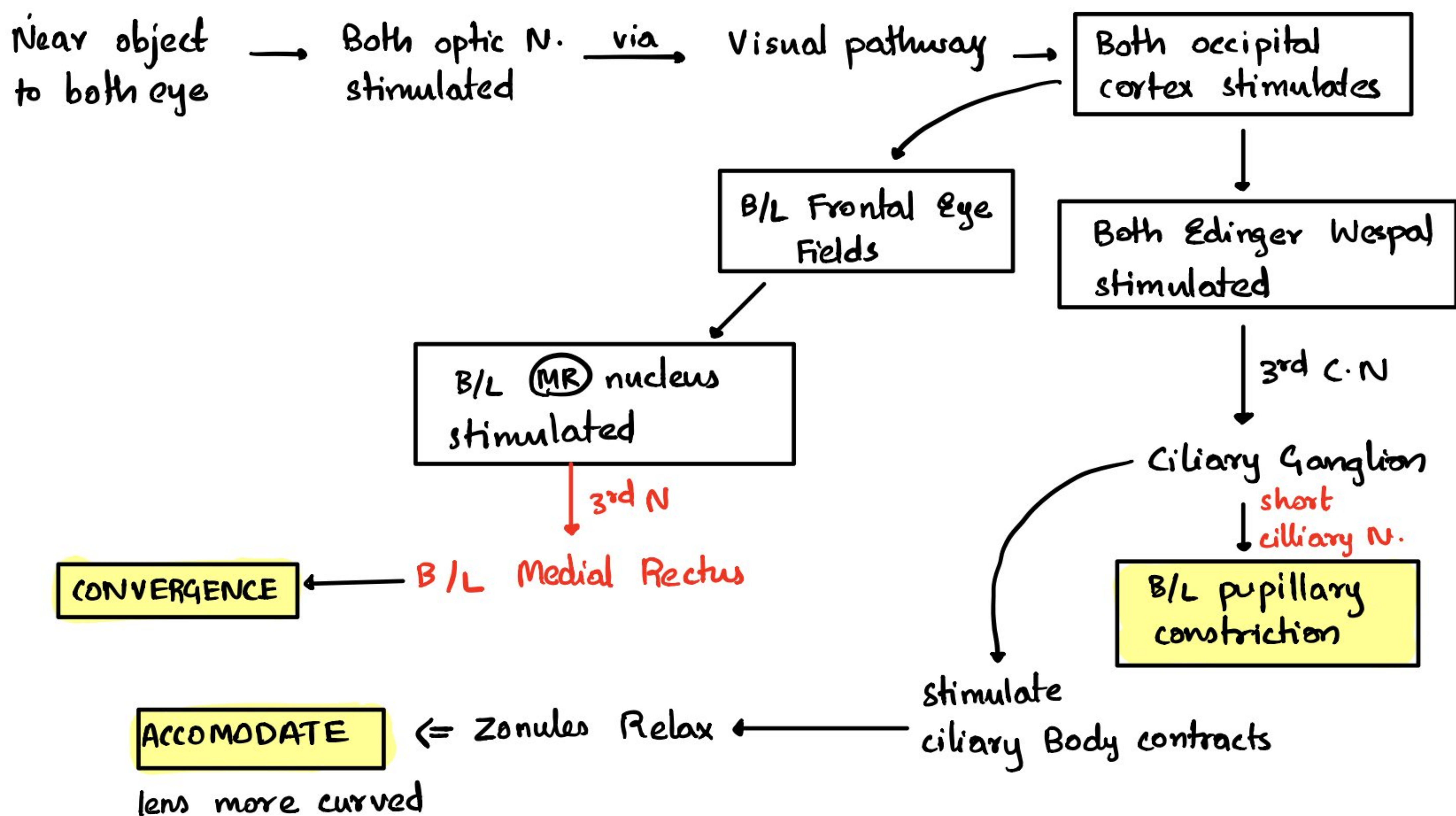


Q: A 67yr old woman finally decides to get glasses, about thirty year after she actually needed them, and goes to an optometrist for an eye glass prescription. Her eye exam reveals some unexpected findings. Her pupil are very small. When a pen light is shone on to her eyes, neither pupil contracts. However, when the optometrist brings his fingers towards her face, both pupils constrict dramatically. What is this pupillary response called?

- a) Anisocoria
- b) Adie's pupil
- c) Ophthalmoplegia interna
- d) Argyll Robertson**



- * Horizontal thickness = (↑)
- * Equatorial thickness = (↓)
- * Anterior chamber depth = (↓)

Accommodation [is for near vision] → Nil after IOL cataract surgery



So glasses are given for near vision

Light Near Dissociation

Argyll Robertson } small constricted pupil

- Accommodation Reflex present
- Lesion @ Pretectal Nucleus
- Dorsal midbrain Syndrome

{ aka PARINAUD Syndrome }

causes: • TABES DORSALIS → Neurosyphilis [classical]

- Myotonic Dystrophy
- DM-1 (MC)

Causes of small and Dilated Pupils:

Causes of small pupils:

- Pontine H'mage
- Horner's Syndrome
- Old age
- Argyll Robertson pupil
- Drugs and poisons (opiates, organophosph)

Causes of dilated pupils

- Holmes - Adie (myotonic) pupil
- 3rd N. palsy
- Atropine
- Carbon monoxide
- Ethylene glycol

Tonic pupil: slow response to light & near

Holmes Adie Pupil

Dilated pupil

Δsis & Rx: Diluted Pilocarpine (0.125%) lesion @ Ciliary Ganglion

* accommodation fibres are spared

↓
(MC) ⇒ Idiopathic

K/cause ⇒ Viral

Dilated pilocarpine used becoz of Denervation hypersensitivity (upreg. of Receptors)

ANISOCORIA:

(R)

○

⊙

(L)

⊙ } ANISOCORIA

"best appreciated in light condition"

⊙

Dark

Hutchinsons Pupil:

- Pupil constricted $\xrightarrow{f/b}$ Pupil dilated and remains dilated
- Occurs in **UNCUS HERNIATION** $\rightarrow$ initially irritation of 3rd N: Pupil constricts $\downarrow f/b$
stretching and damage: Pupil dilates $\downarrow f/b$ of 3rd N.

ADIE's Tonic pupil

- Post ganglionic lesion affecting Ciliary ganglion
- Dilated pupil $\rightarrow$ d/t sympathetic takeover
- Sluggish response to near & light reflex
- **Wormiform movement of pupil**
- Post ganglion parasympathetic fibres affected $\rightarrow$ No ACh release
Sphincture $\rightarrow$ No ACh $\rightarrow$ Denervation Hypersensitivity to ACh receptor (starvation)
 $\swarrow$ **CONstrict** Pilocarpine $\{ < < < 0.01\% \}$

ADIE's Syndrome (Adie Holmes's Syndrome):

- 1) ADIE's pupil
- 2) B/L Deep tendon reflex absent
 $\swarrow$ knee jerk $\ominus$ $\searrow$ Ankle jerk $\ominus$
- 3) Young ♀

ANISOCORIA: Pupil size are unequal

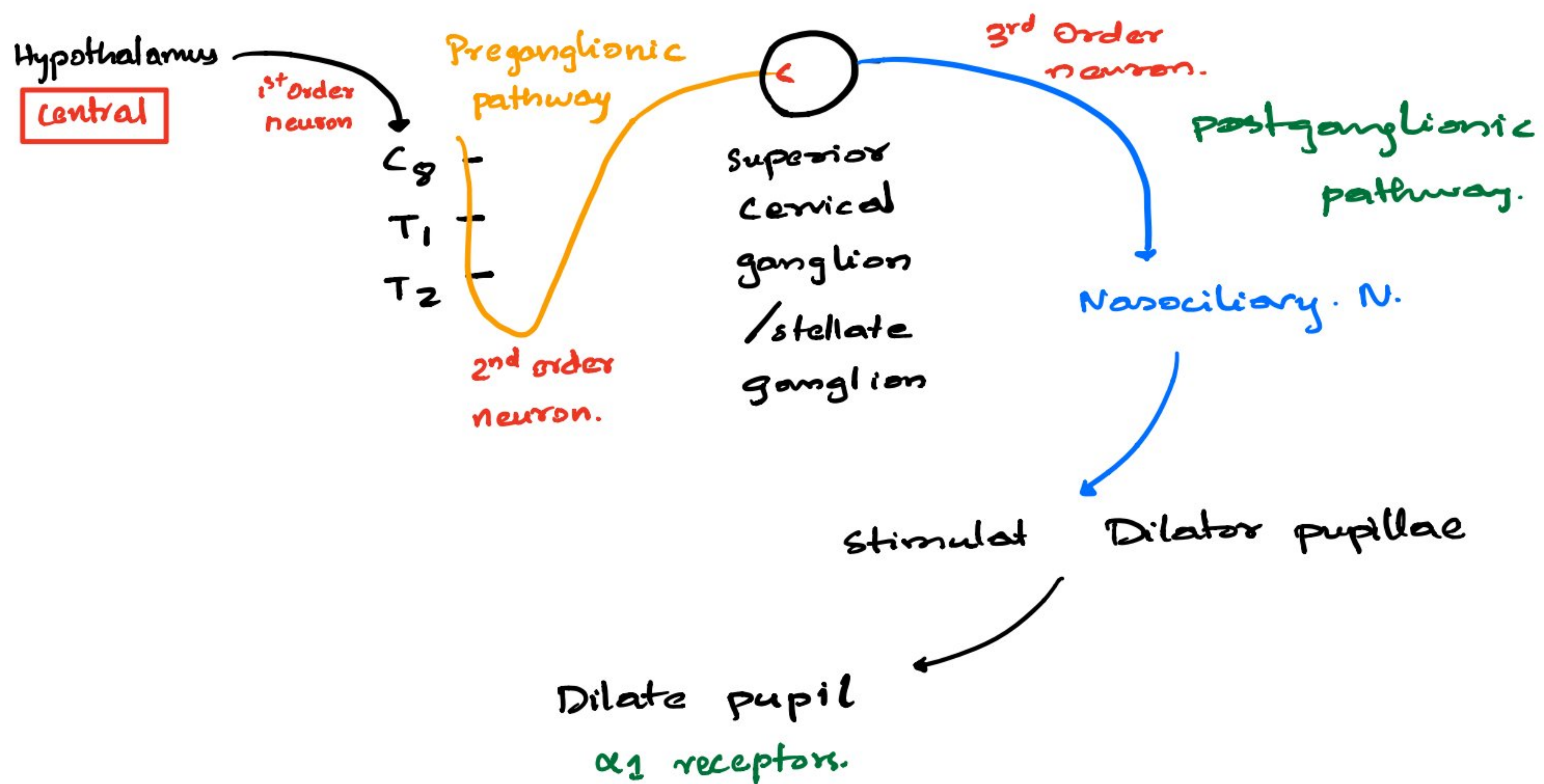
- **(MCC)** $\rightarrow$ Physiological (20%) aka simple / essential anisocoria
 $\downarrow$
diff upto 0.3mm.
- Pathological anisocoria ($>0.3\text{mm}$)
diff.
- Anisocoria $\rightarrow$ in dim light $\uparrow$ ses $\Rightarrow$ Sympathetic system } HORNER'S sx

$\textcircled{R}$


$\textcircled{L}$


"DILATATION Lag"

SYMPATHETIC PATHWAY:

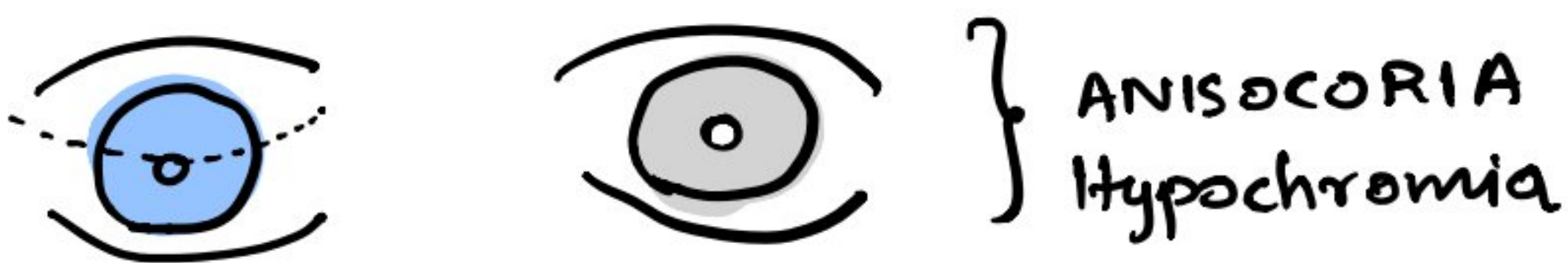


HORNER'S SYNDROME ← ↓se in sympathetic pathway

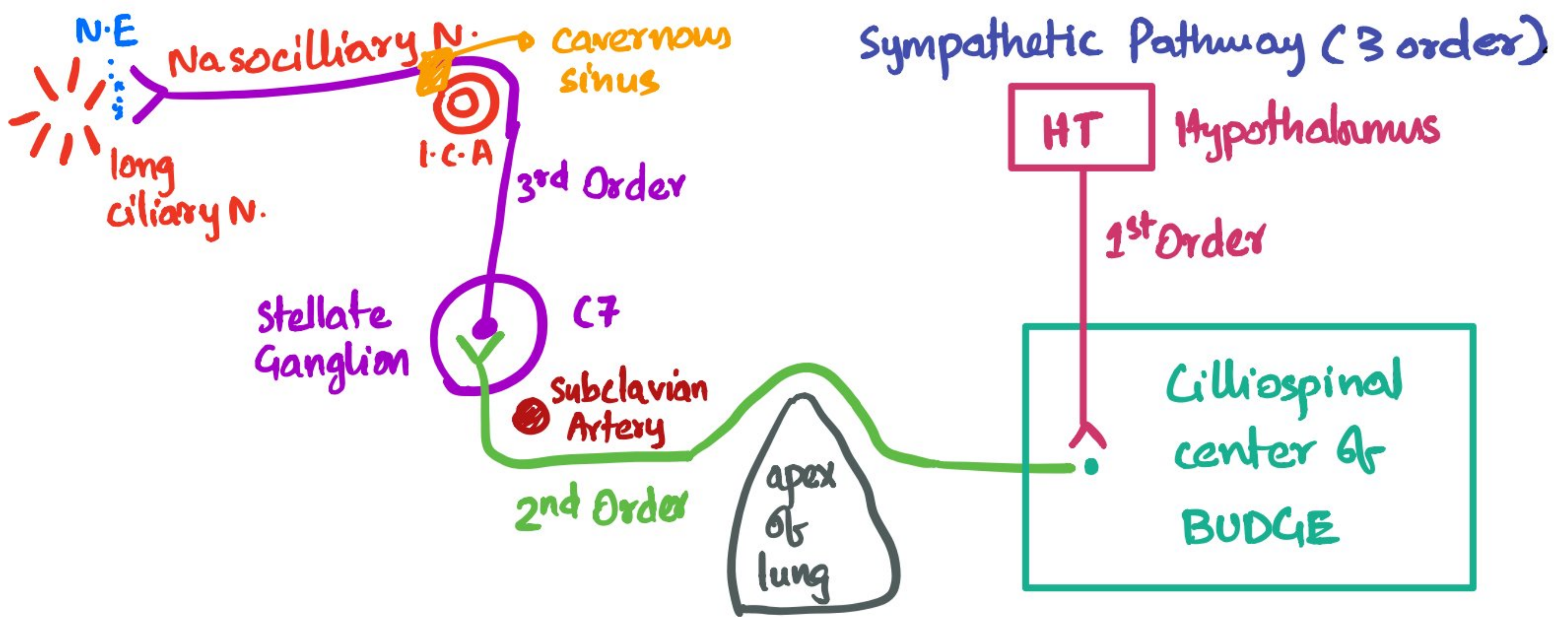
- 1) Miosis [**dilator muscle**]
- 2) Partial ptosis [mild] = Muller's muscle.
- 3) Psuedoenophthalmus → **sunken eyeball & inferior tarsal muscles**
- 4) HETEROCHROMIA IRIS & s/i congenital Horner's Syndrome
- 5) Anhidrosis
- 6) Loss of ciliospinal reflex → **on neck stroking, pupil will not dilate anymore**

Congenital Horner's Sx

Hypochromia Iris → Involved side



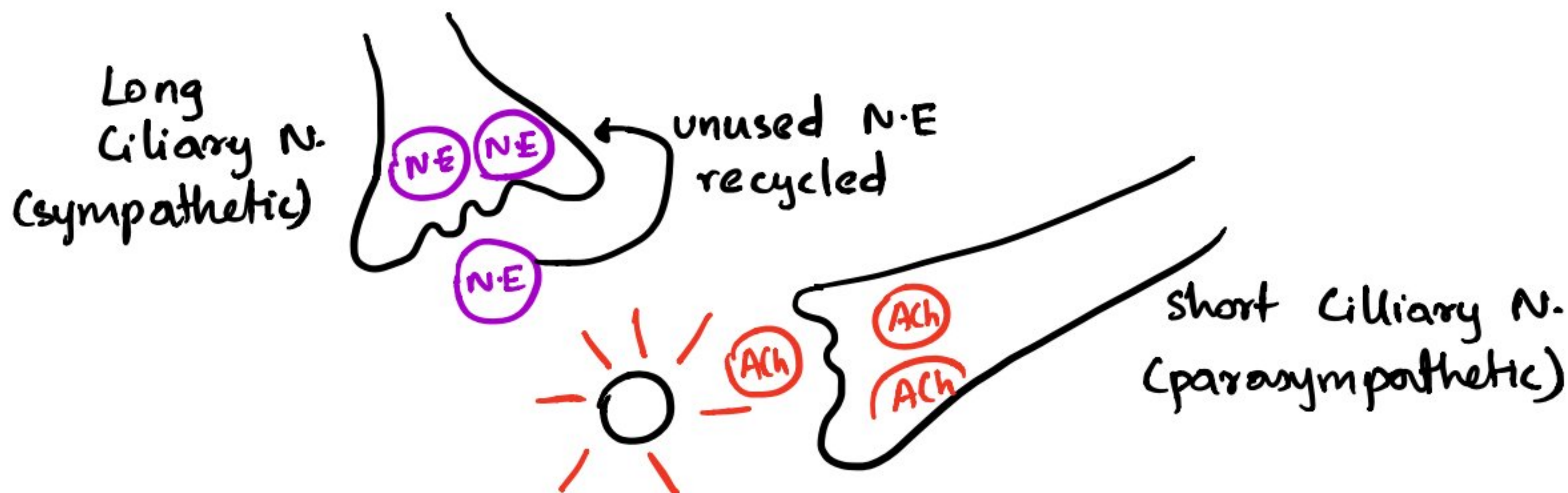
Cause ⇒ NCC migration failure } Melanocytes as origin : NCC
 ⊕
 Sympathetic stimulation



Types of Horner's Sx

- 1) Lesion of Hypothalamus (1st order) ⇒ **CENTRAL Horner's Syndrome**
5, 7, 8, 9, 10, 11 CN affect ← Lateral Medullary syndrome / Wardenburg Sx
- 2) Lesion of 2nd order neuron ⇒ **Preganglionic Horner's Syndrome [Pancoast TUMOUR]**
- 3) Lesion of 3rd order neuron ⇒ **Post-ganglionic Horner's Syndrome {ICA Aneurysm}**

(N) Physiology @ NMJ [dilator pupillae]



Confirmation test ⇒ **COCAINE eyedrop** → prevents reuptake of released N.E.

Localisation test ⇒ **Hydroxy amphetamine** → stimulation of N.E. release

Pre-ganglionic lesion ⇒ Stimulus is Absent

NT ⊕ load in presynaptic vesicle

Post-ganglionic lesion → No N.T @ presynaptic vesicle



	(N) / central Horner	Preganglionic Horner	Post-ganglionic
Cocaine for confirmation	Dilatation of pupil Abundant NT → ↑↑↑ dilatation / suboptimal dilation.	No dilatation No N.T released @ NMJ all are loaded in vesicle	No dilatation Scarcity of N.T

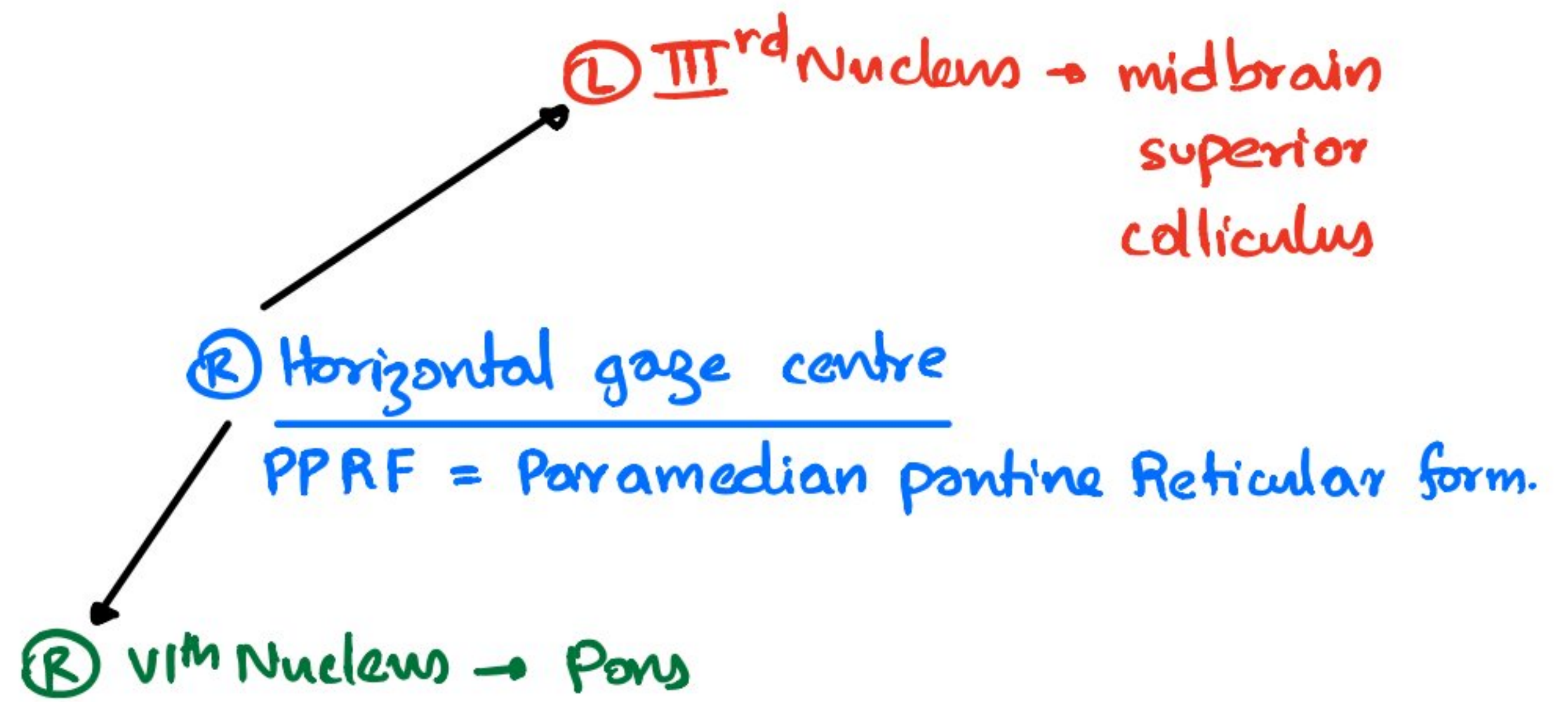
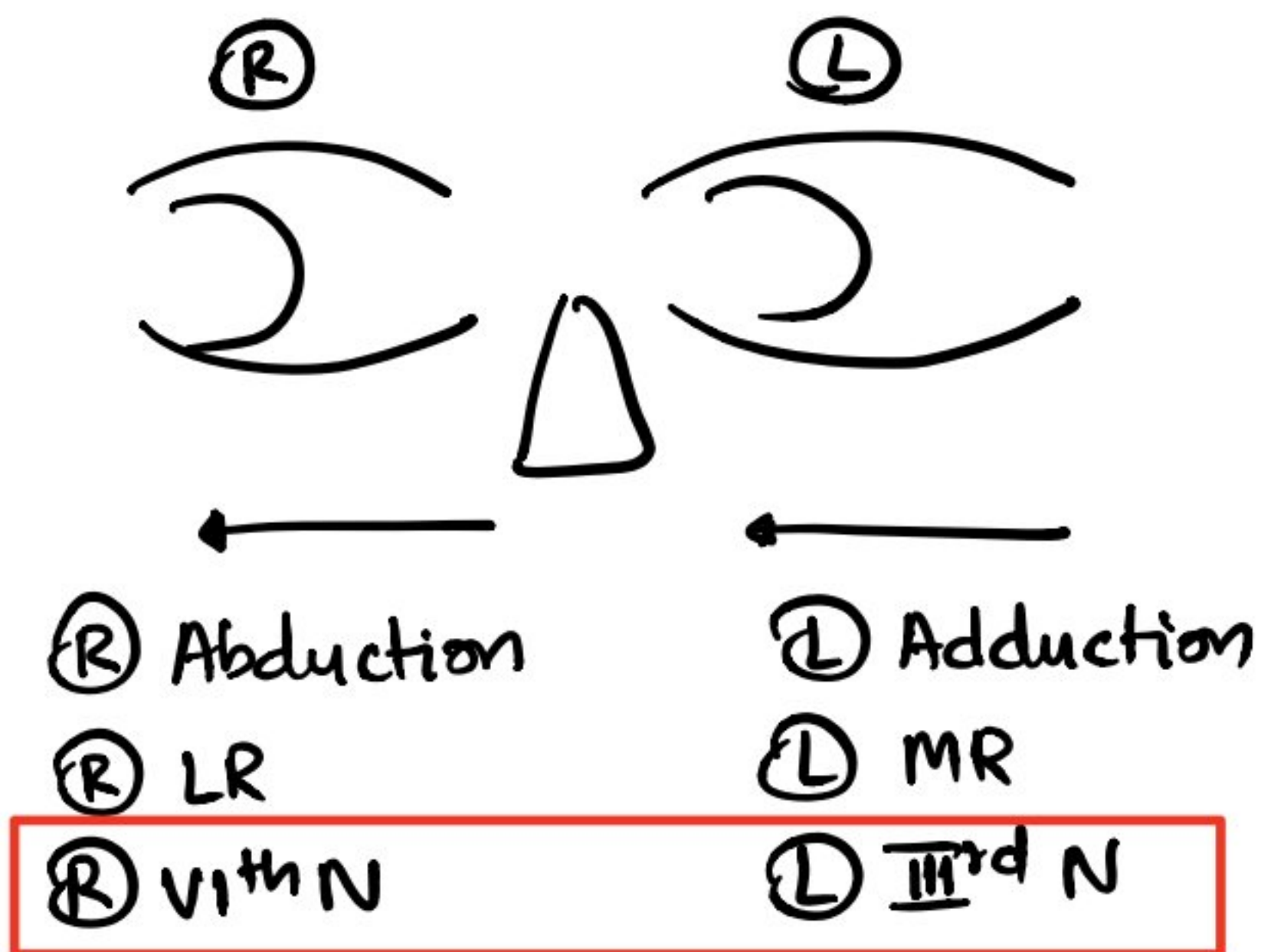
Hydroxi amphetamine	Not done /suboptimal dilatation	Dilatation of pupil stimulus provided by HA for release of N.T	No dilatation
Cause		PANCOAST TUMOUR	ICA aneurysm
Investigation		CT - CHEST	MR Angiography

DISORDERS OF HORIZONTAL GAZE

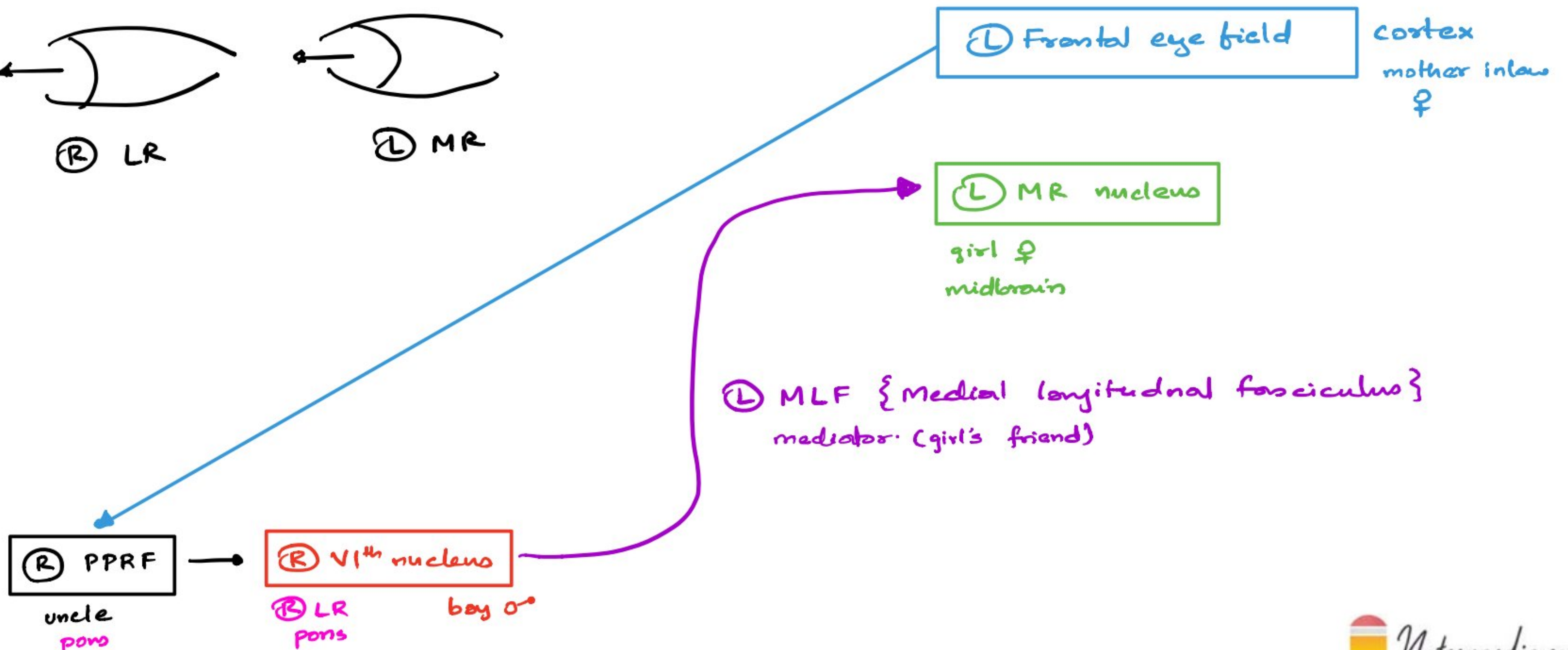
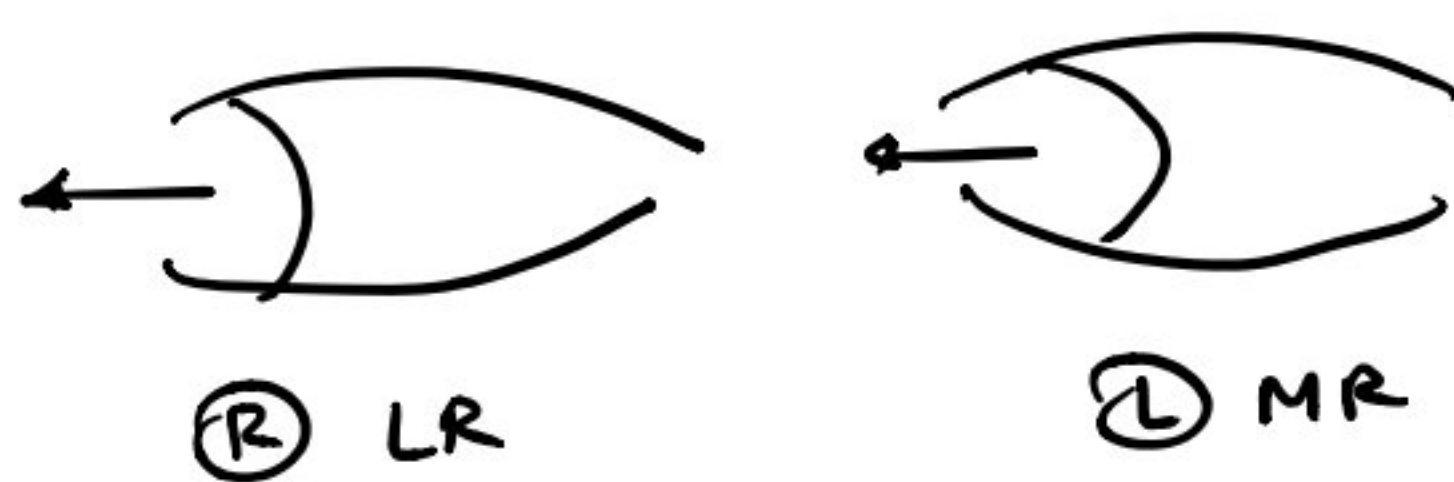
Ⓡ Gaze : Dextro version
Ⓡ Binocular conjugate
movt.

Ⓛ Gaze : Levo version

Ⓡ gaze



RIGHT GAZE



Lesion of PPRF $\rightarrow$ I/L horizontal gaze palsy

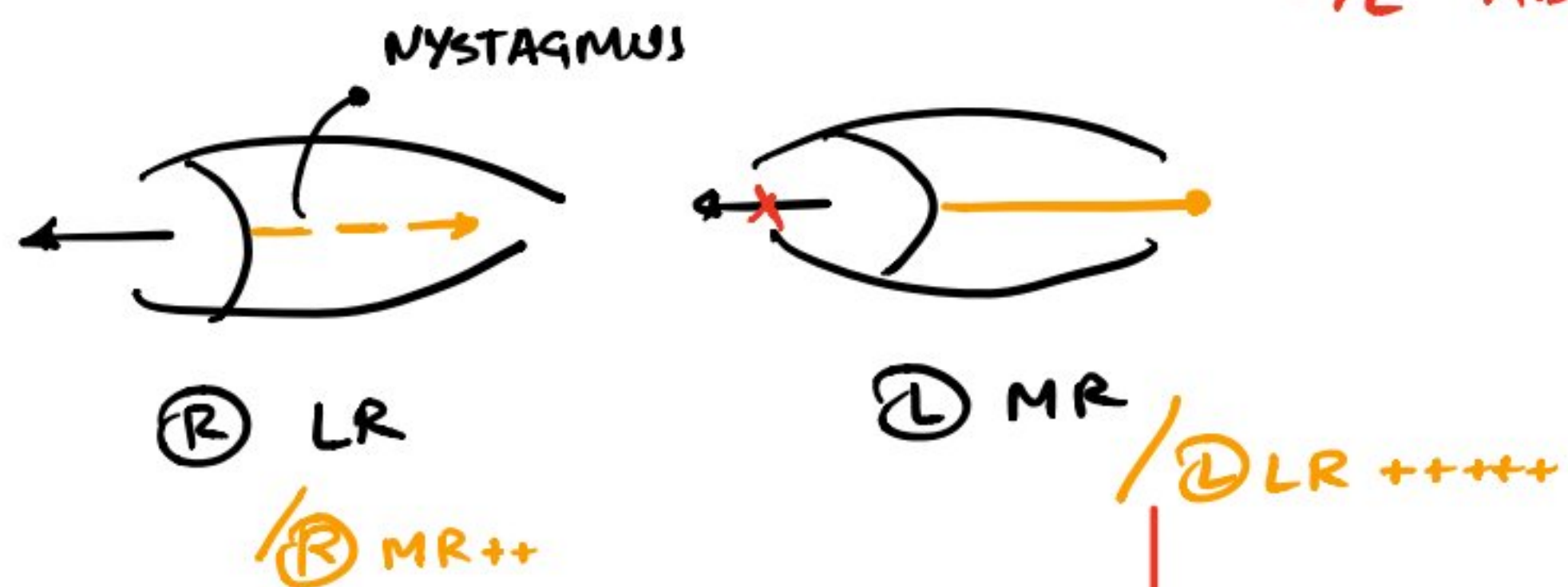


Pontine Lesion $\rightarrow$ eye look away from the lesion

Lesion @ FEF lesion → 4/L horizontal gaze palsy
1/L skew deviation

* Frontal lobe lesion → eye look towards the lesion

Lesion @ MLF → I/L Adduction deficit
C/L Abduction & nystagmus.



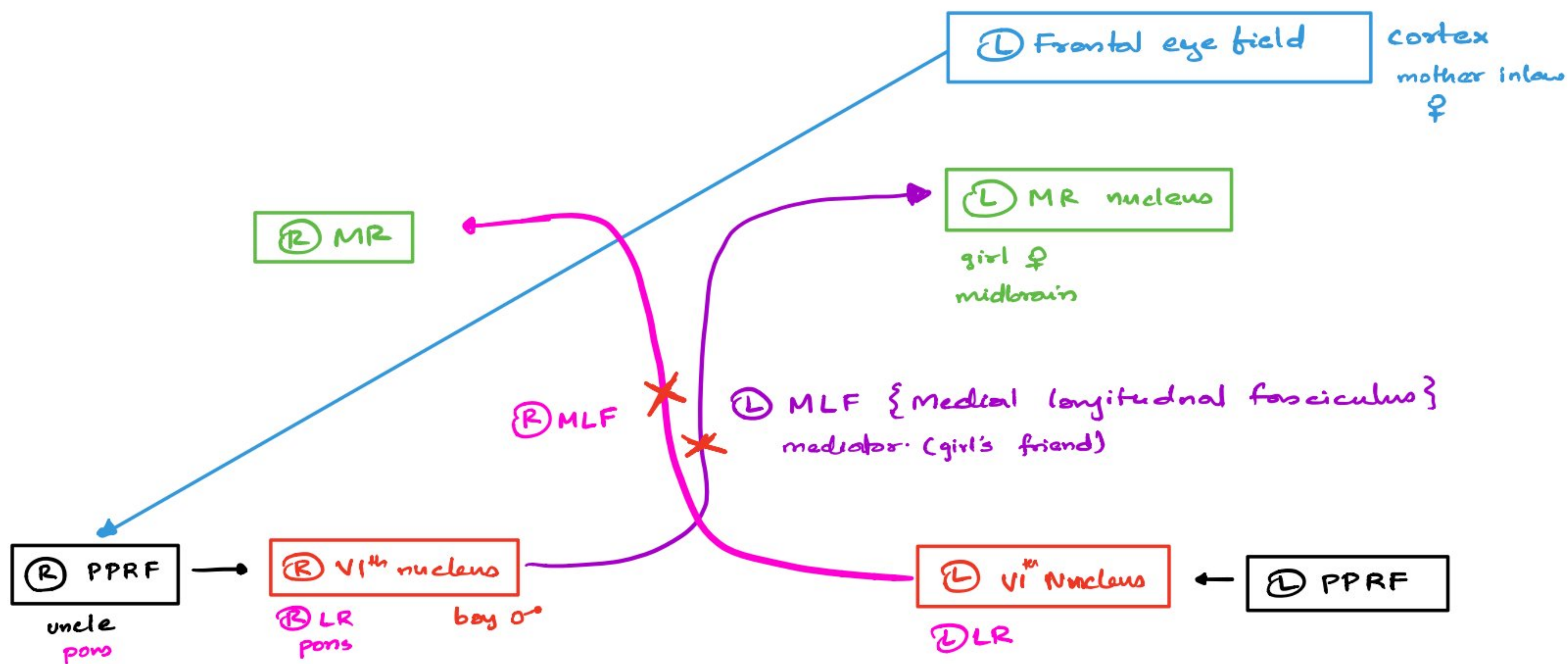
d/t Hering law of equal innervation of yolk muscle.

② Abduction & Nystagmus. ③ Adduction deficit

INTERNUCLEAR OPHTHALMOPLÉGIA (INO)

MCC: multiple sclerosis

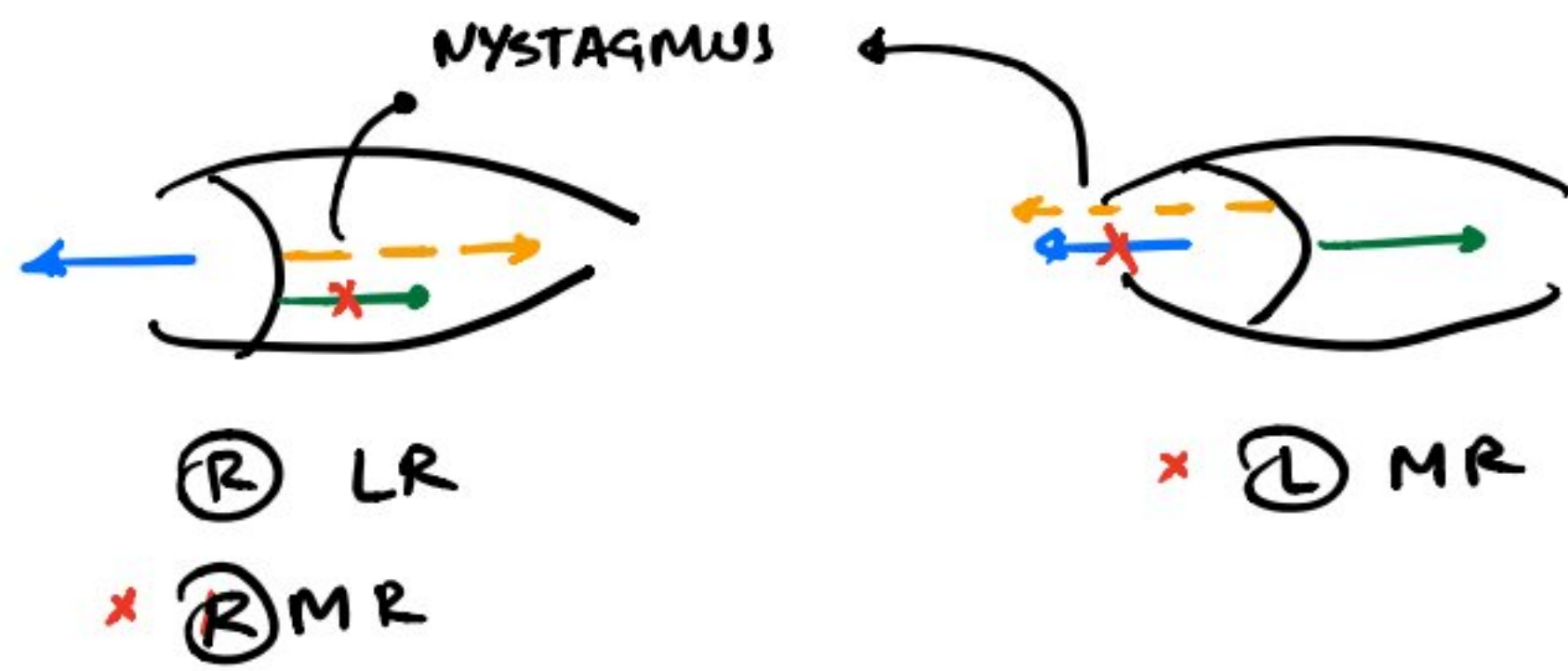
Lesion at B/L MLF:



Ⓡ Abduction & nystagmus

Ⓛ Adduction deficit

Ⓡ GAZE



Ⓛ GAZE

Ⓡ Adduction deficit

Ⓛ Abduct & Nystagmus.

B/L Adduction deficit

+

B/L divergence ?
Nystagmus.

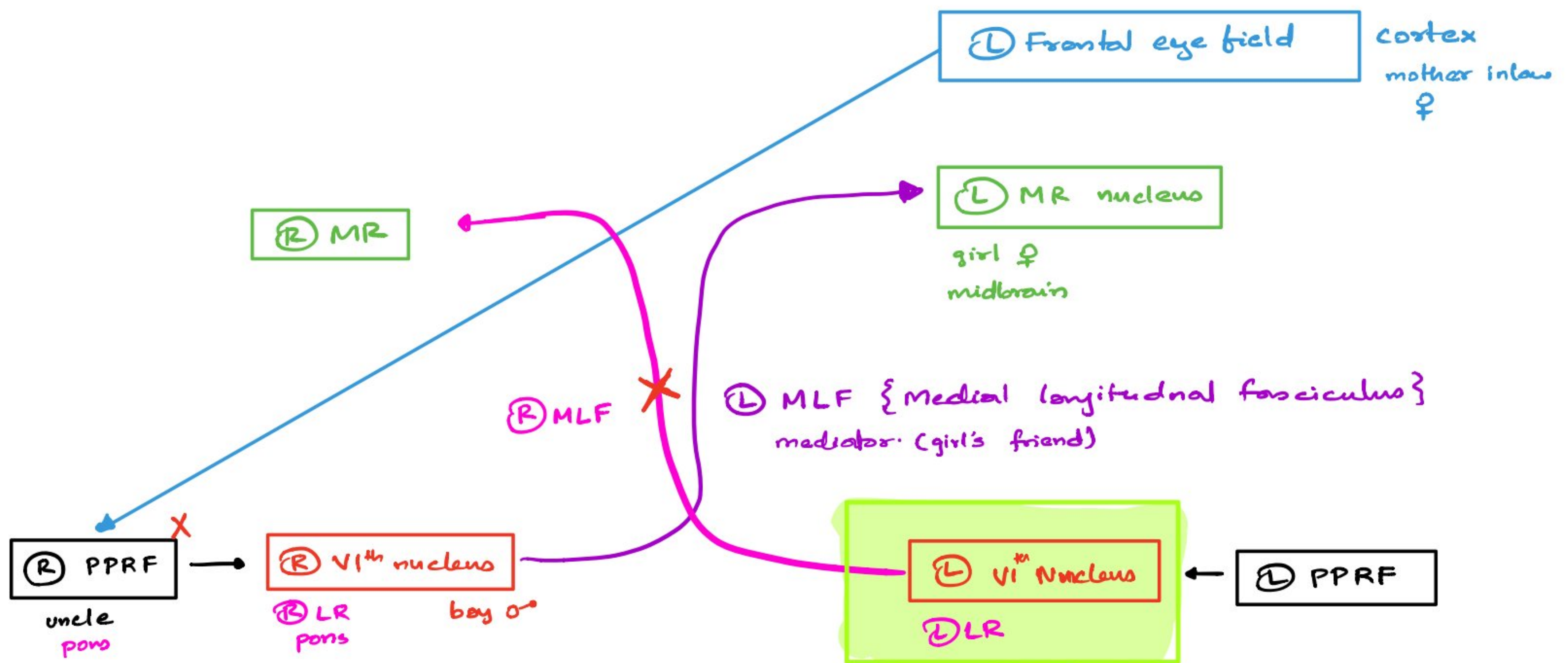


wall eyed



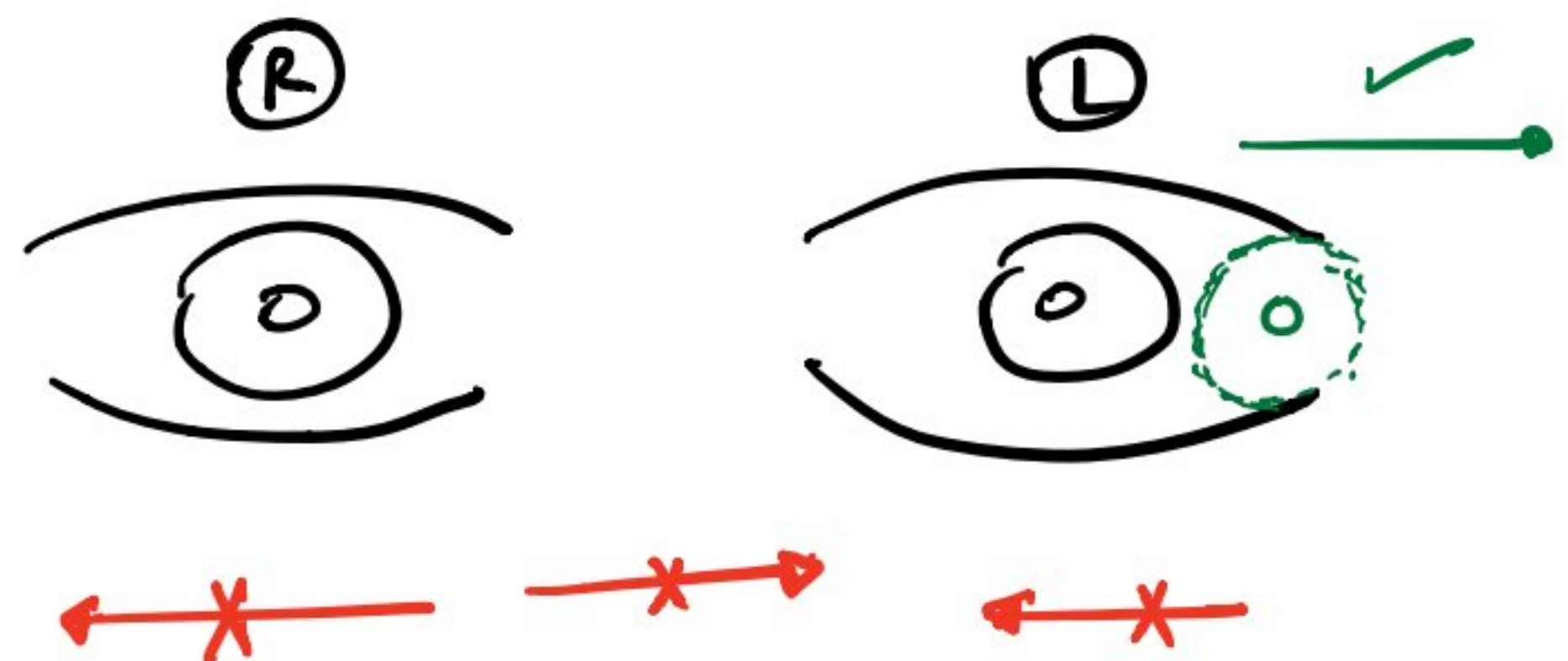
B/L INO ⇒ **WEBINO Syndr.**

ⓂCC = M.S.



Ⓡ PPRF lesion → Ⓡ gaze palsy

Ⓡ MLF lesion → Ⓡ Adduction deficit



ONE and a HALF syndrome

↳ C/L Abduction spared.
C/L divergent squint.

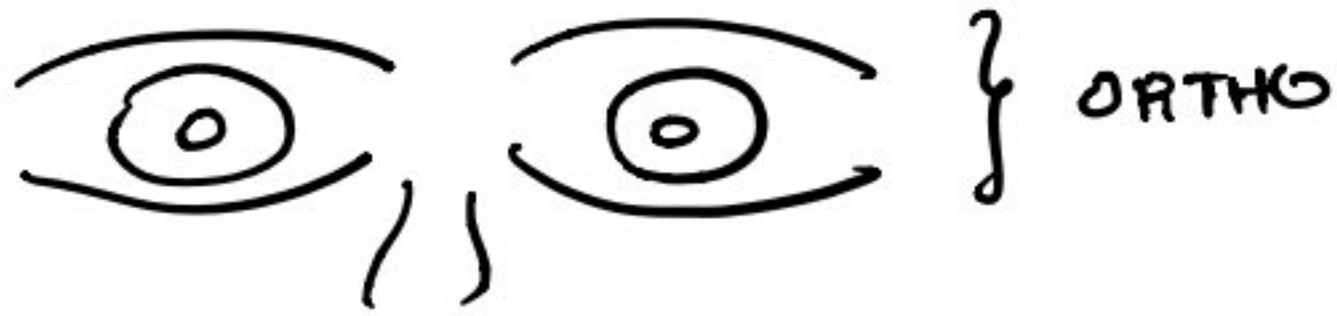
Lesion of PPRF & MLF → C/L abduction is spared
& 1 & 1/2 syndrome

{ I/L complete loss of horizontal movt.
C/L adduction deficit }

Q. on attempted left gaze, the (R) eye does not move beyond midline. and the (L) eye has a nystagmus. Where is the site of lesion?

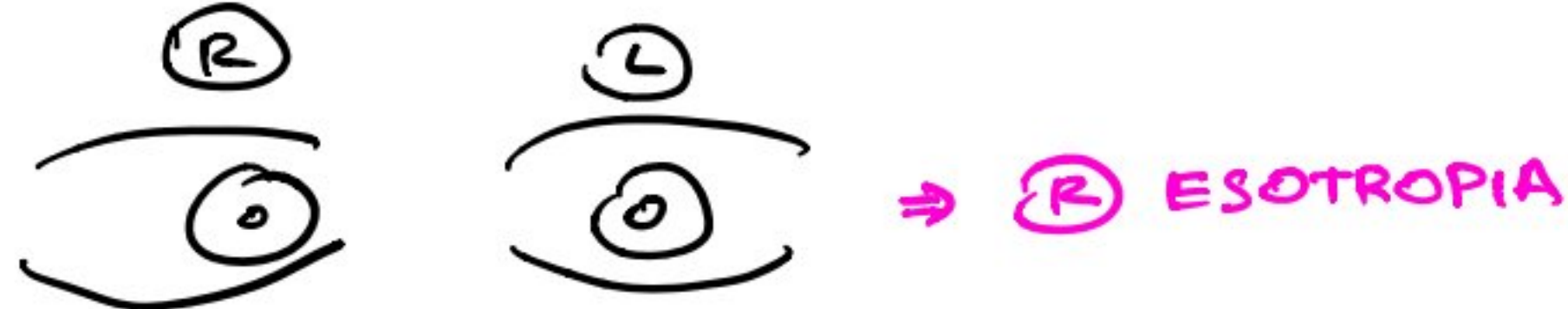
sol. (R) MLF lesion $\rightarrow$ (R) INO

SQUINT / Strabismus :



ESO = convergent squint

EXO = divergent squint



⑫ convergent series $\{ \text{manifest} \}$

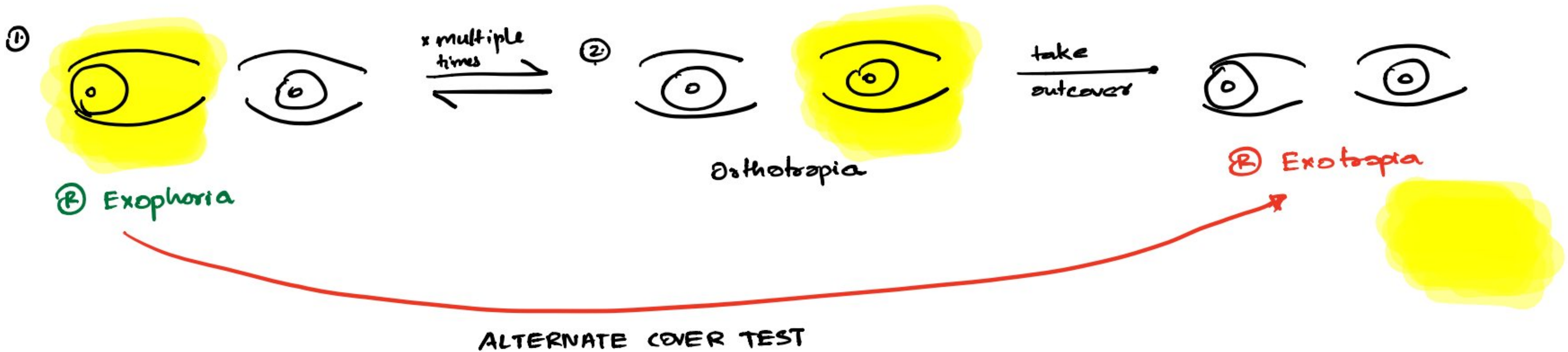


⑫ divergent squint



Redressal ment.

⇒ corrective movt.



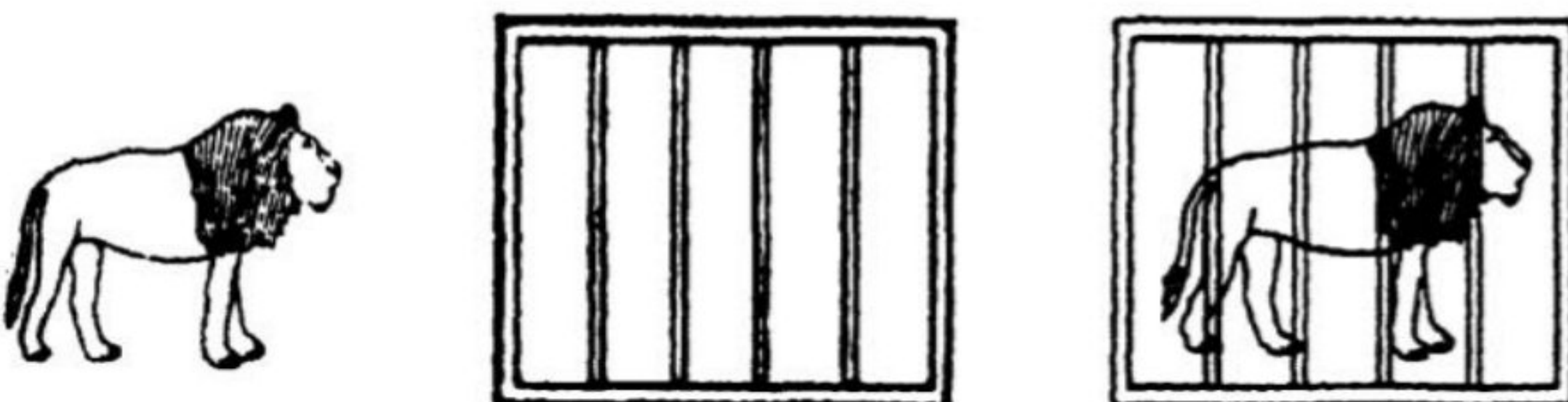
BINOCULAR SINGLE VISION (BSV)

Stages of BSV

I, Simultaneous Macular perception (SMP)

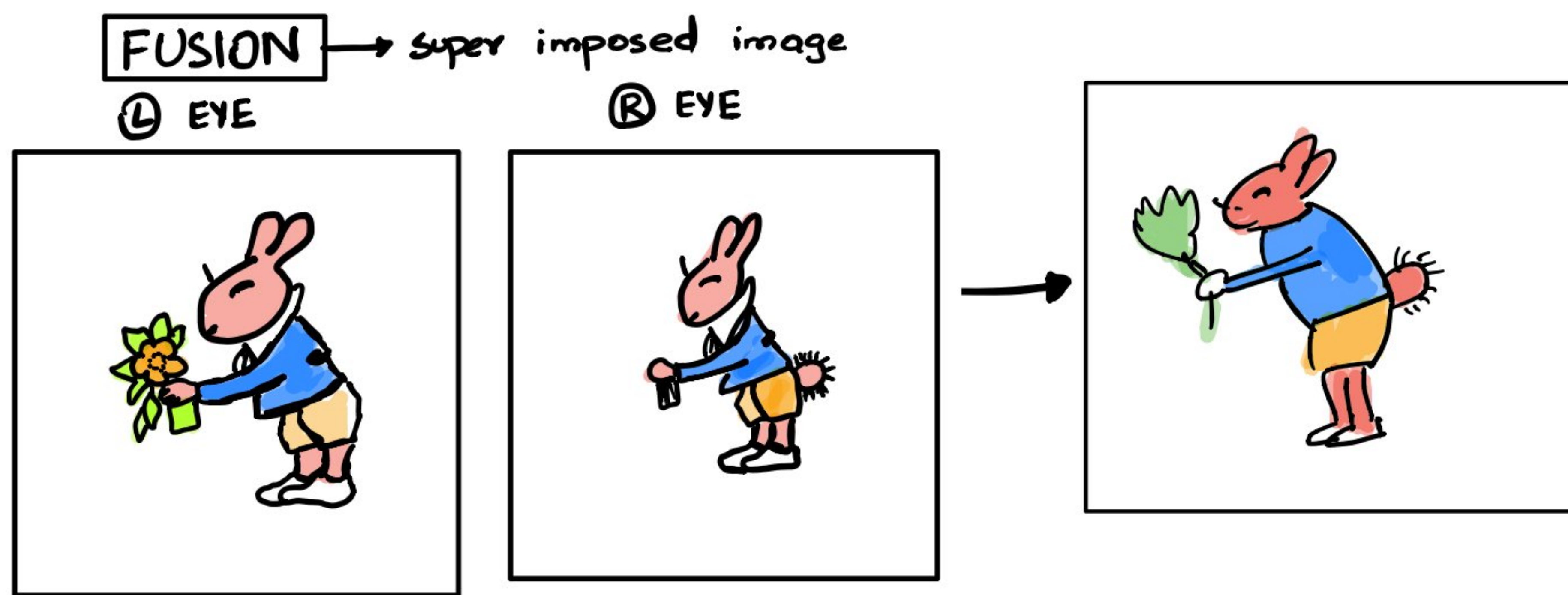
1.First grade (Simultaneous perception)

- ▶ Tested by introducing two dissimilar but not mutually antagonistic pictures, such as lion and a cage.

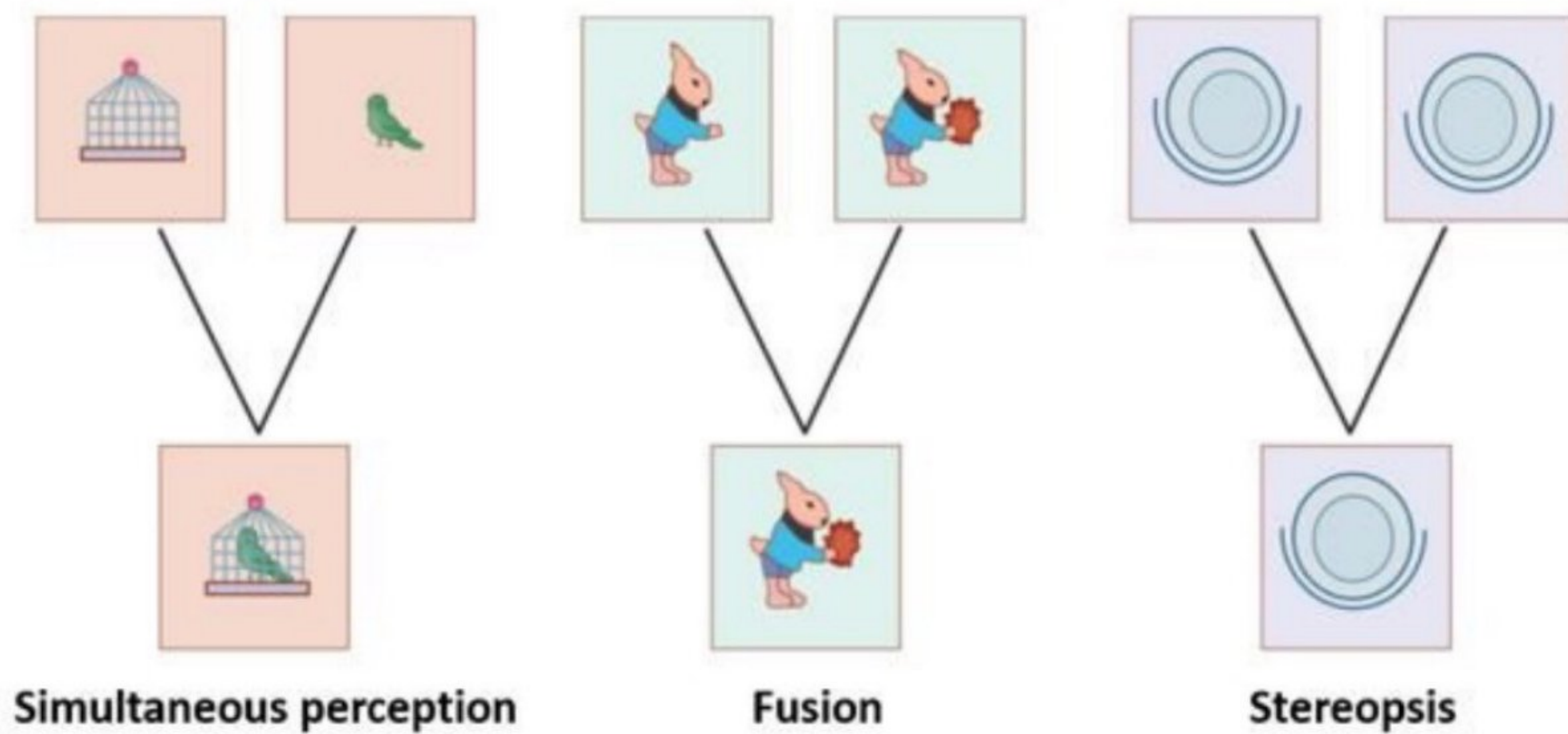
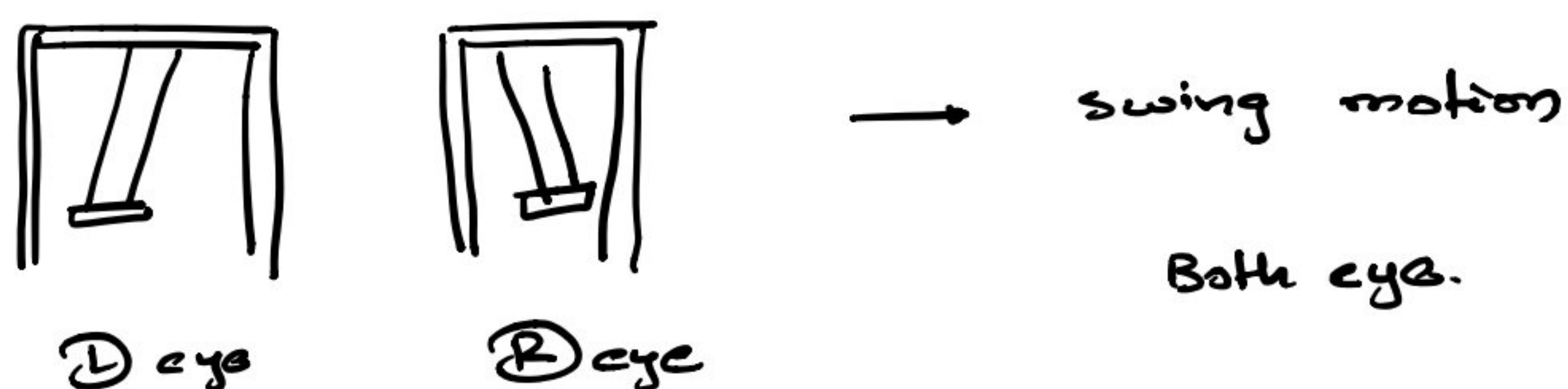


(e) Slides for testing simultaneous macular perception. (Lion and cage.)

II) Fusion



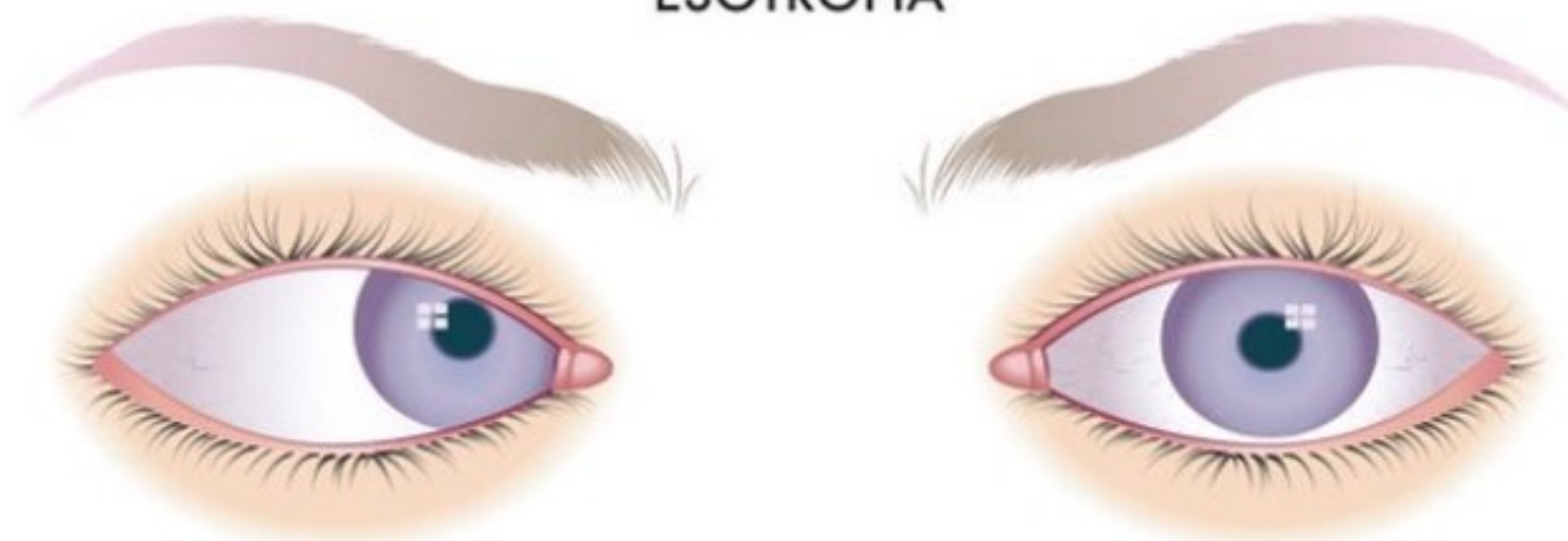
III) STEREOPSIS → Depth perception



EXOTROPIA



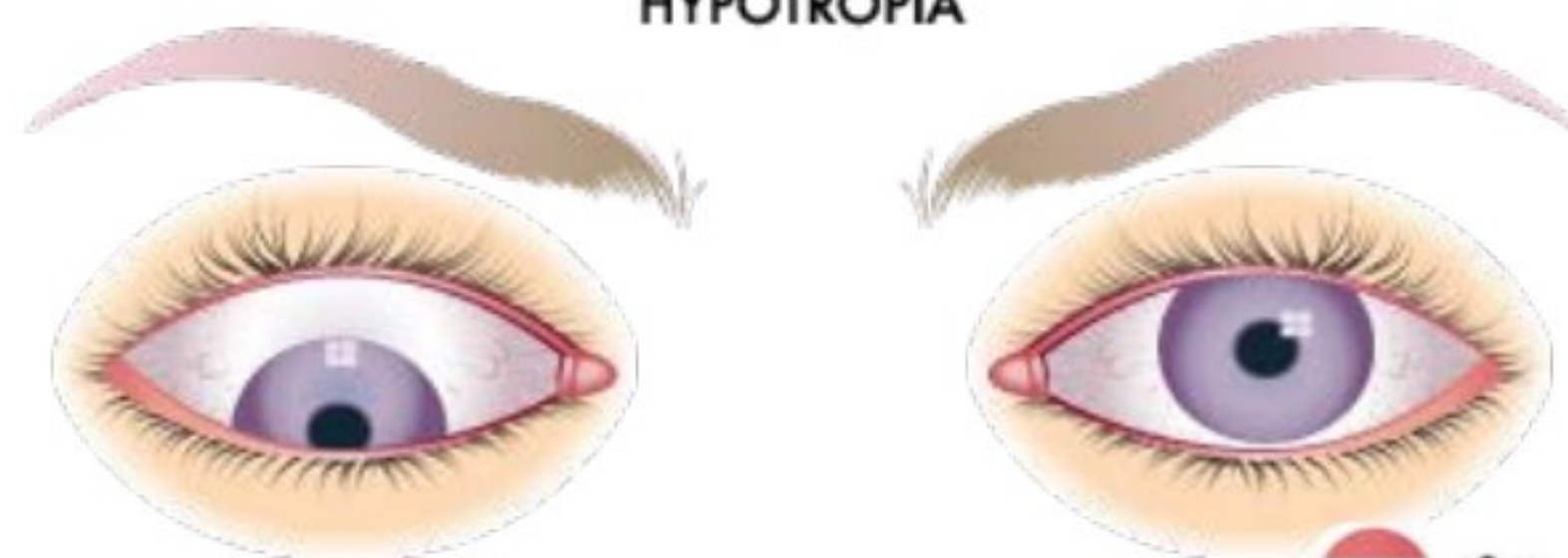
ESOTROPIA



HYPERTROPIA



HYPOTROPIA



① CONVERGENT SQUINT

① ESOTROPIA



d/t ① LR palsy
or Vith No palsy

① Adducted
{ ① MR overacting }

Agonist - Antagonist pair of movement

Adduction

AGONIST : MR

ANTAGONIST : LR (I/L)

① Adduction : Agonist ① MR $\uparrow \oplus$ + Antagonist ① LR $\downarrow \ominus$ } SHERRINGTON'S LAW OF RECIPROCAL INHIBITION

Agonist

Antagonist

pair

MR

LR

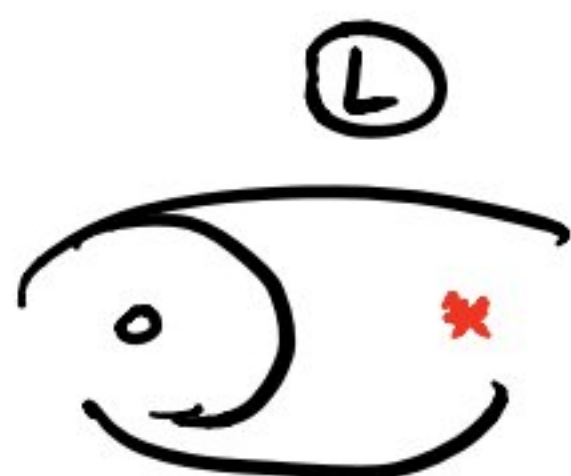
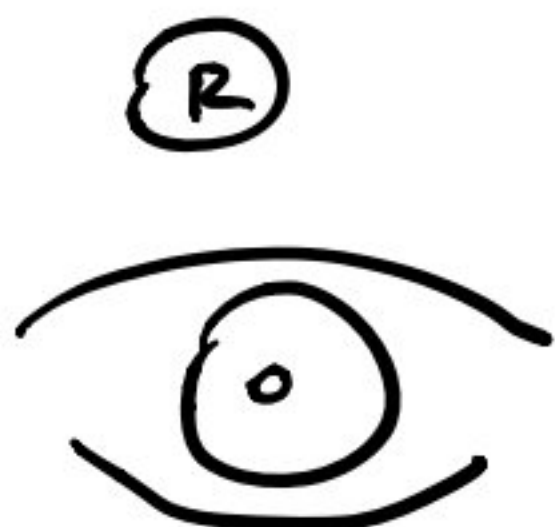
SR

IR

SO

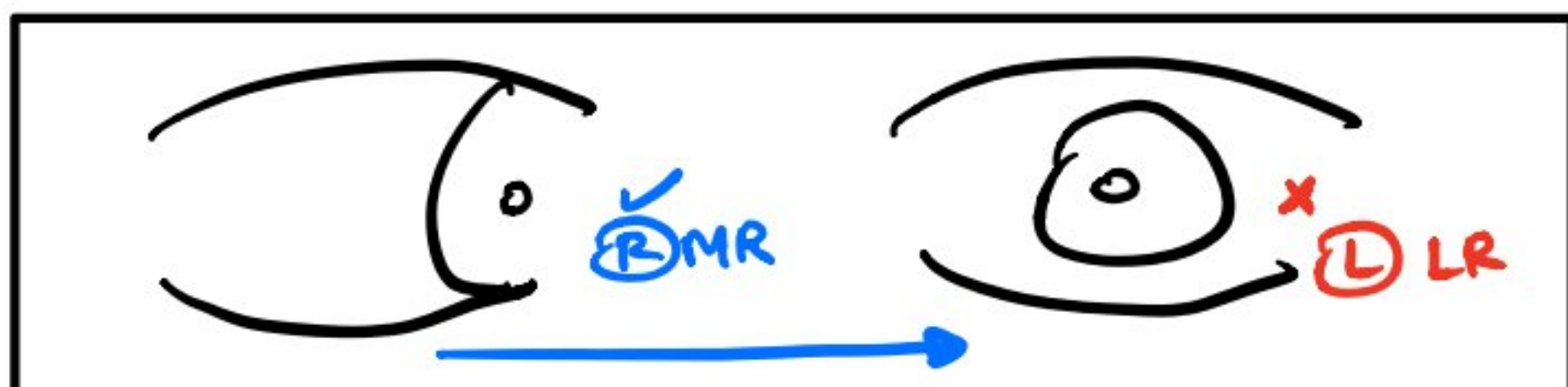
IO

} sherringtons law of Reciprocal inhibition } If one muscle is not working other muscle starts overacting



→ 1° gaze

① LR palsied $\Rightarrow$ ① MR overacting



① gaze

① Abduction deficit.

} MAXM diplopia.

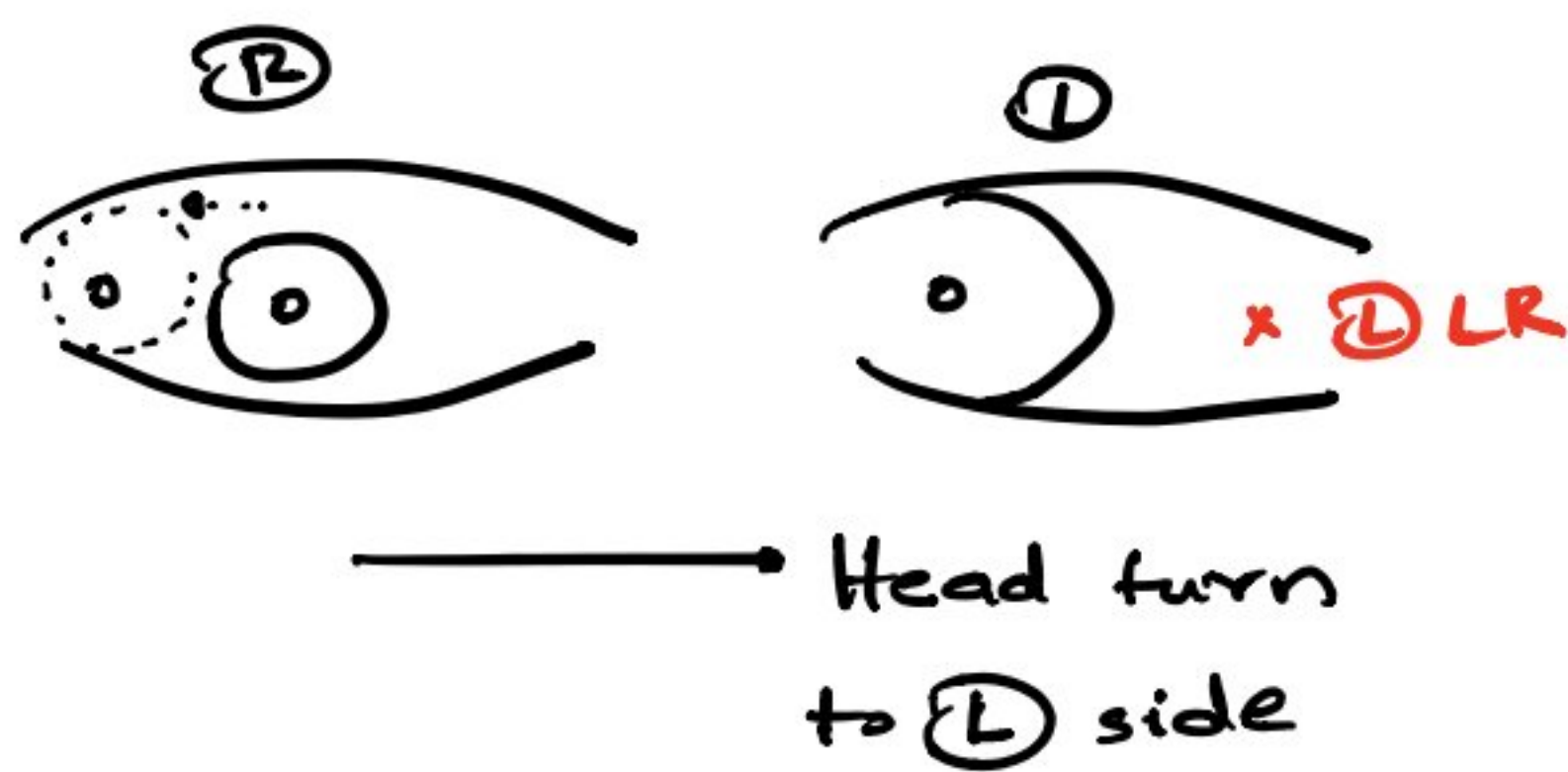


① gaze

NORMAL

SQUINT $\rightarrow$ Binocular diplopia

Compensatory Head posture / anomalous head posture



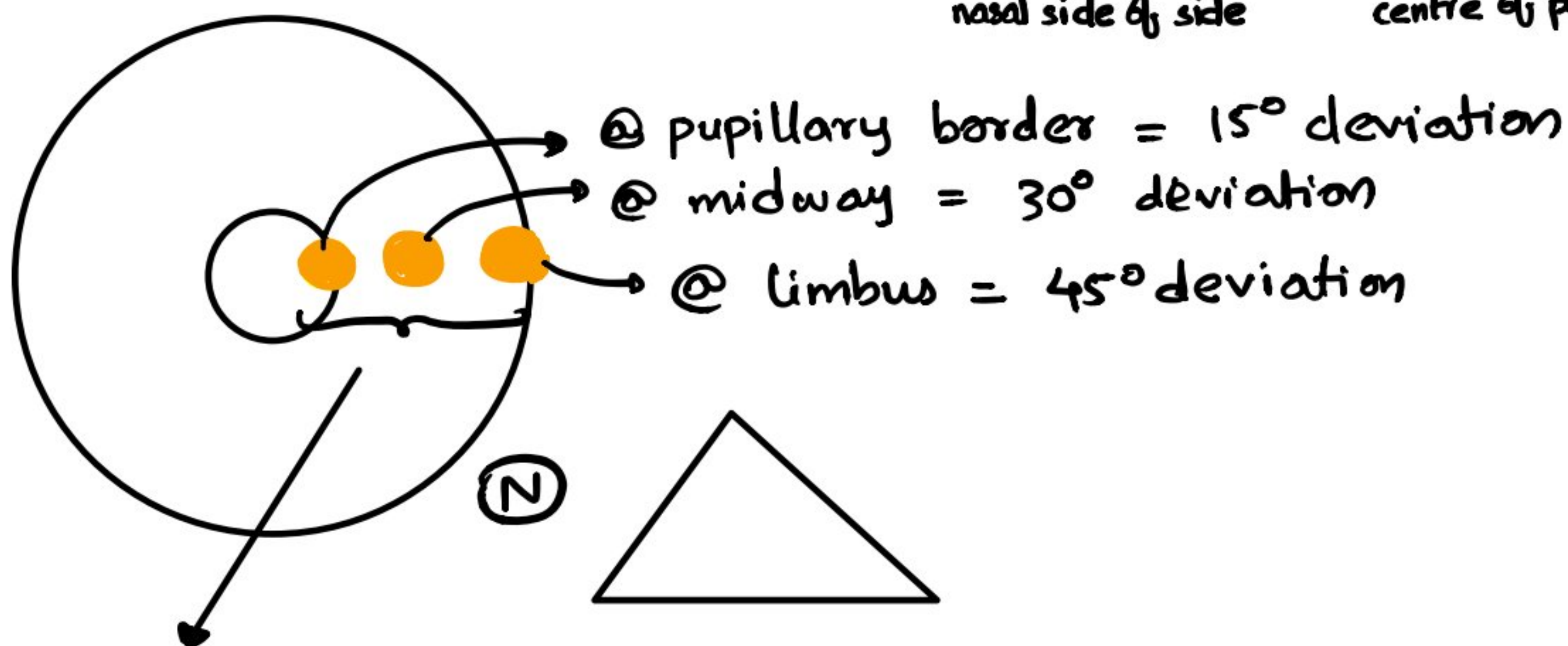
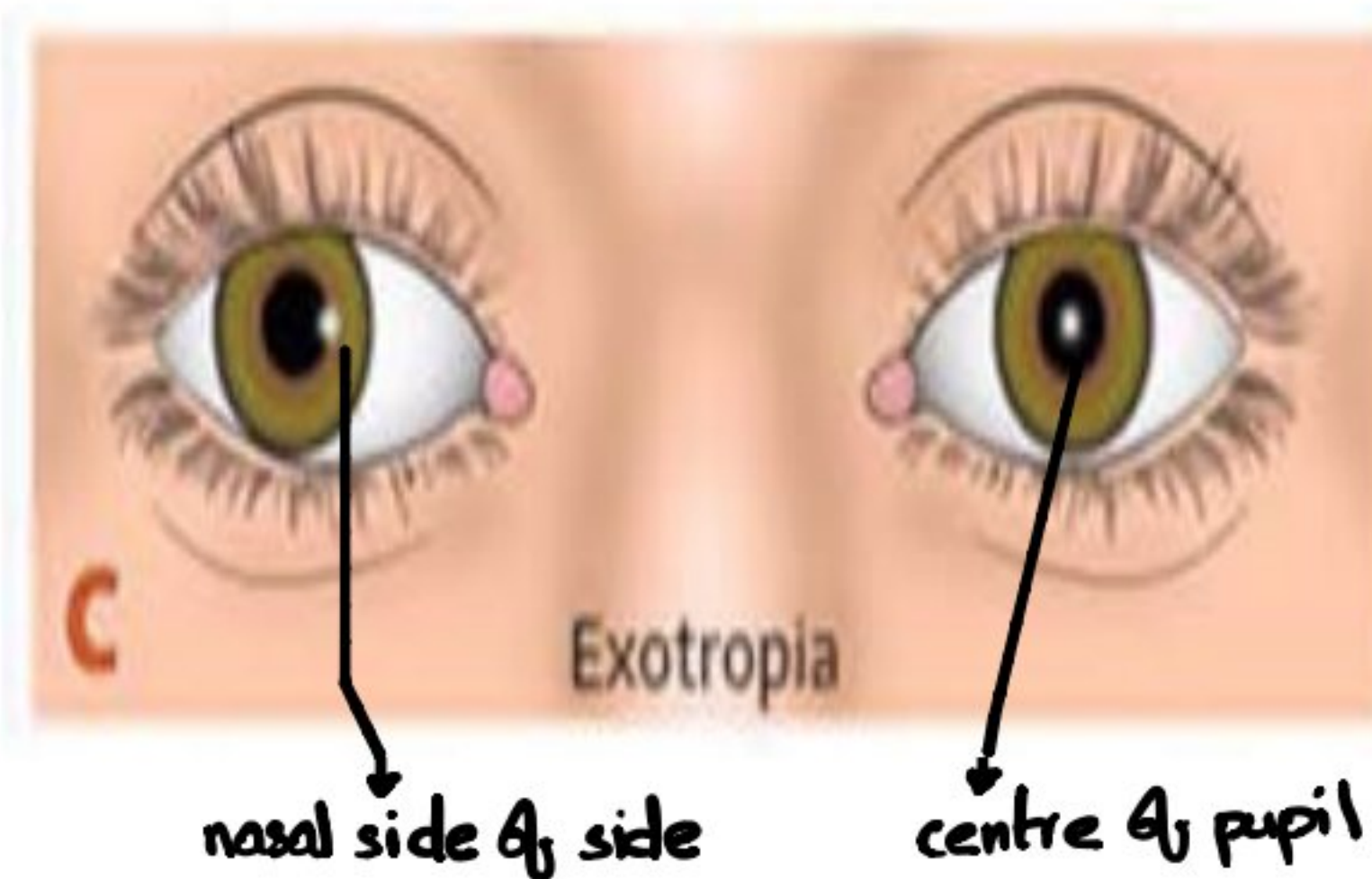
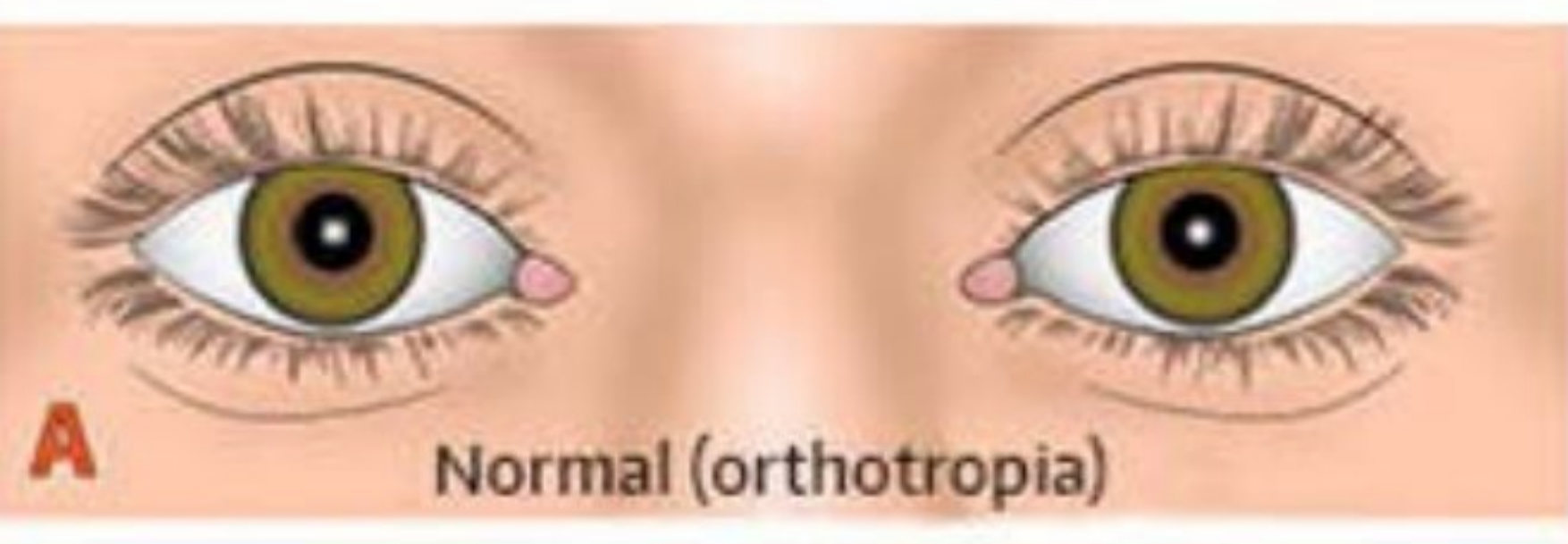
- 1) LR palsy = Head / face turn to some side of lesion
- 2) SO Palsy = Head tilt to opp. shoulder

Angle of Squint:

- 1) Hirschberg's test (using torchlight)
- 2) Prism / Prism bar
- 3) Krinsky's test (Hirschberg + Prism)
- 4) Synaptophore

Unit of angle = Degree or Prism dioptre

$$1^\circ = 2\text{PD}$$

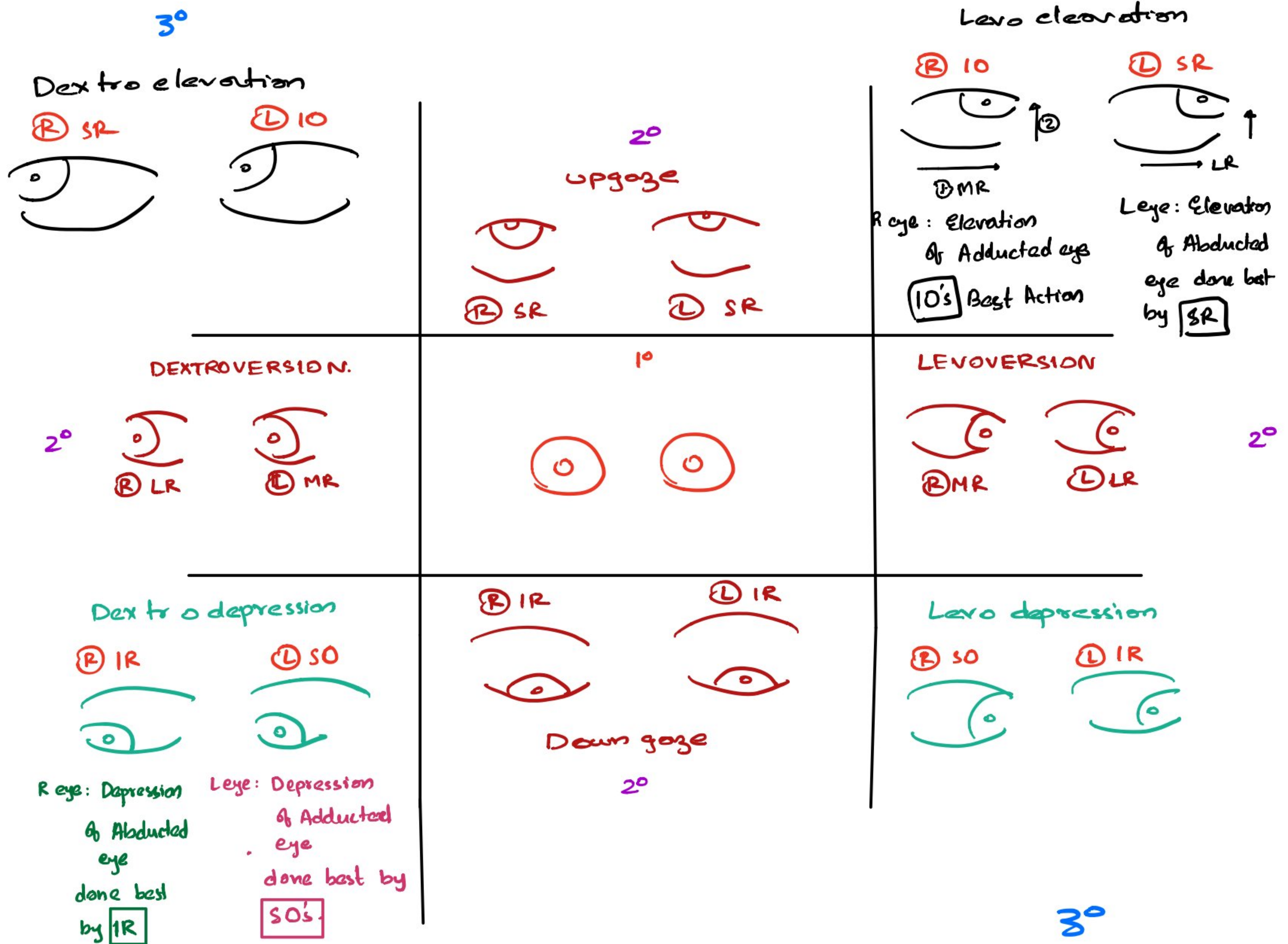


Nasal side = Exotropia

Gazes : Binocular movement

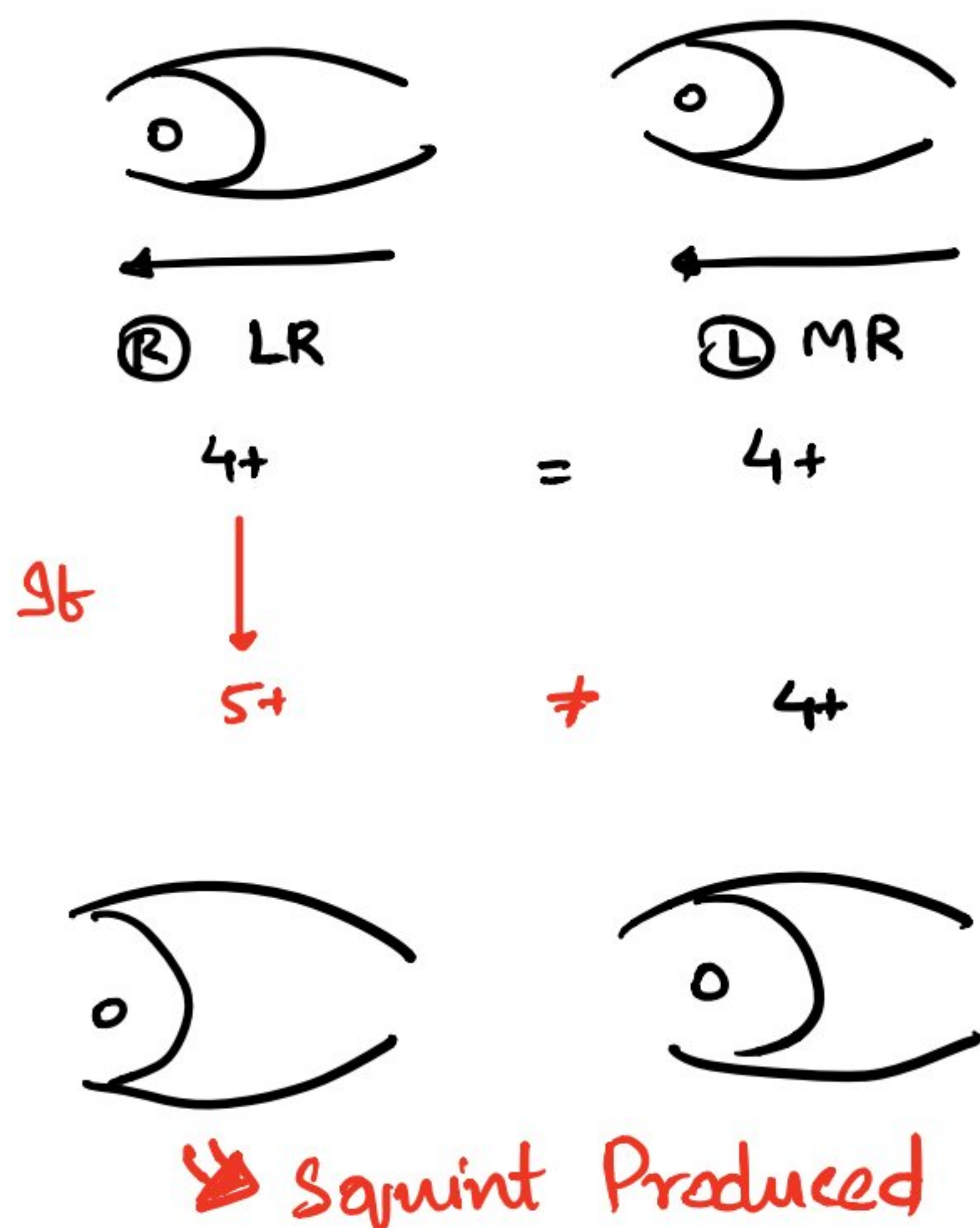
Conjugate: visual axis of two eyes are parallel to each other VERSION movt.

$$\text{Total} = 9 \text{ gazes } \approx 1^\circ + 4(2^\circ) + 4(3^\circ)$$

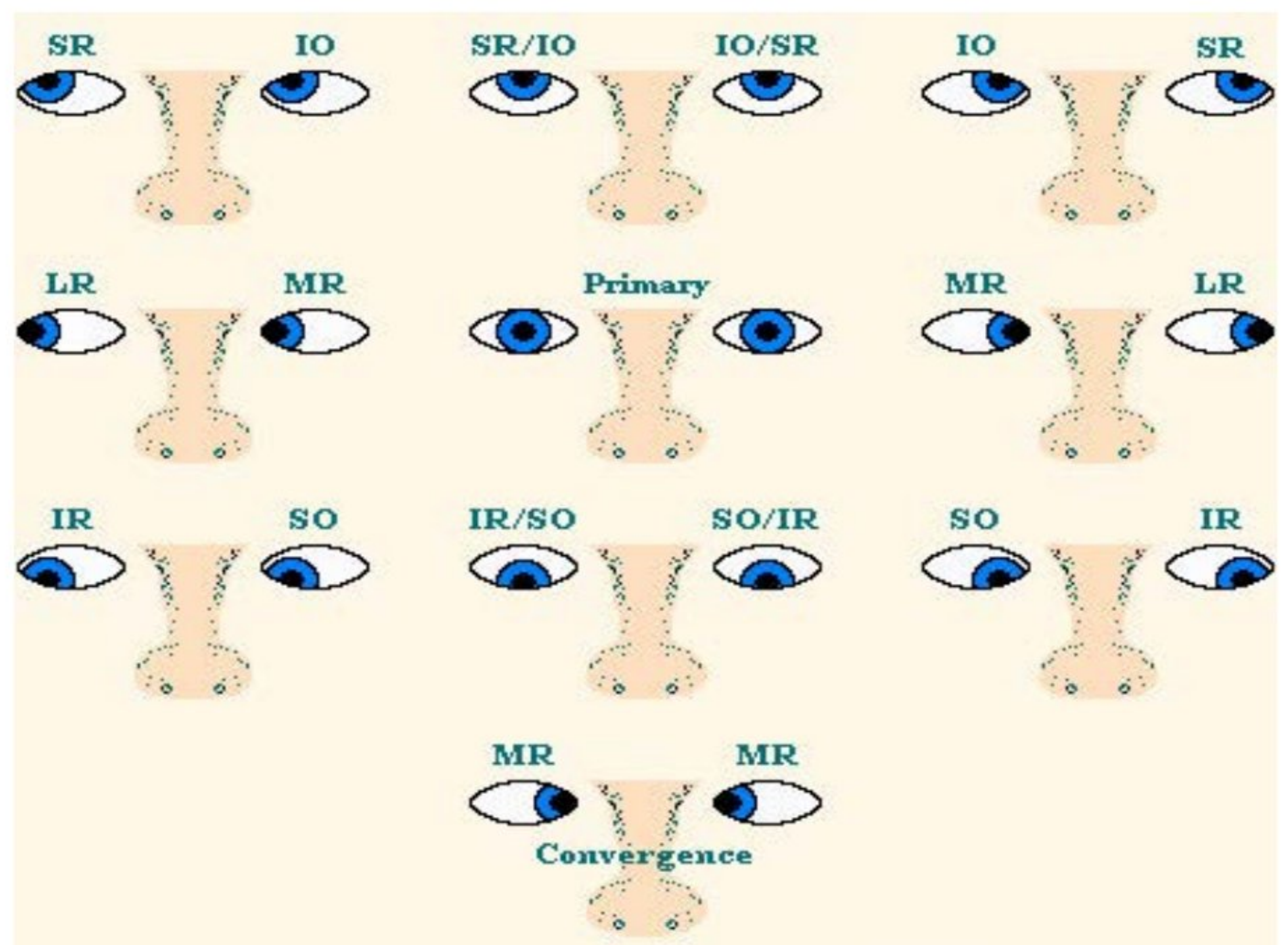


YOLK MUSCLES FOR THEIR RESPECTIVE GAZES

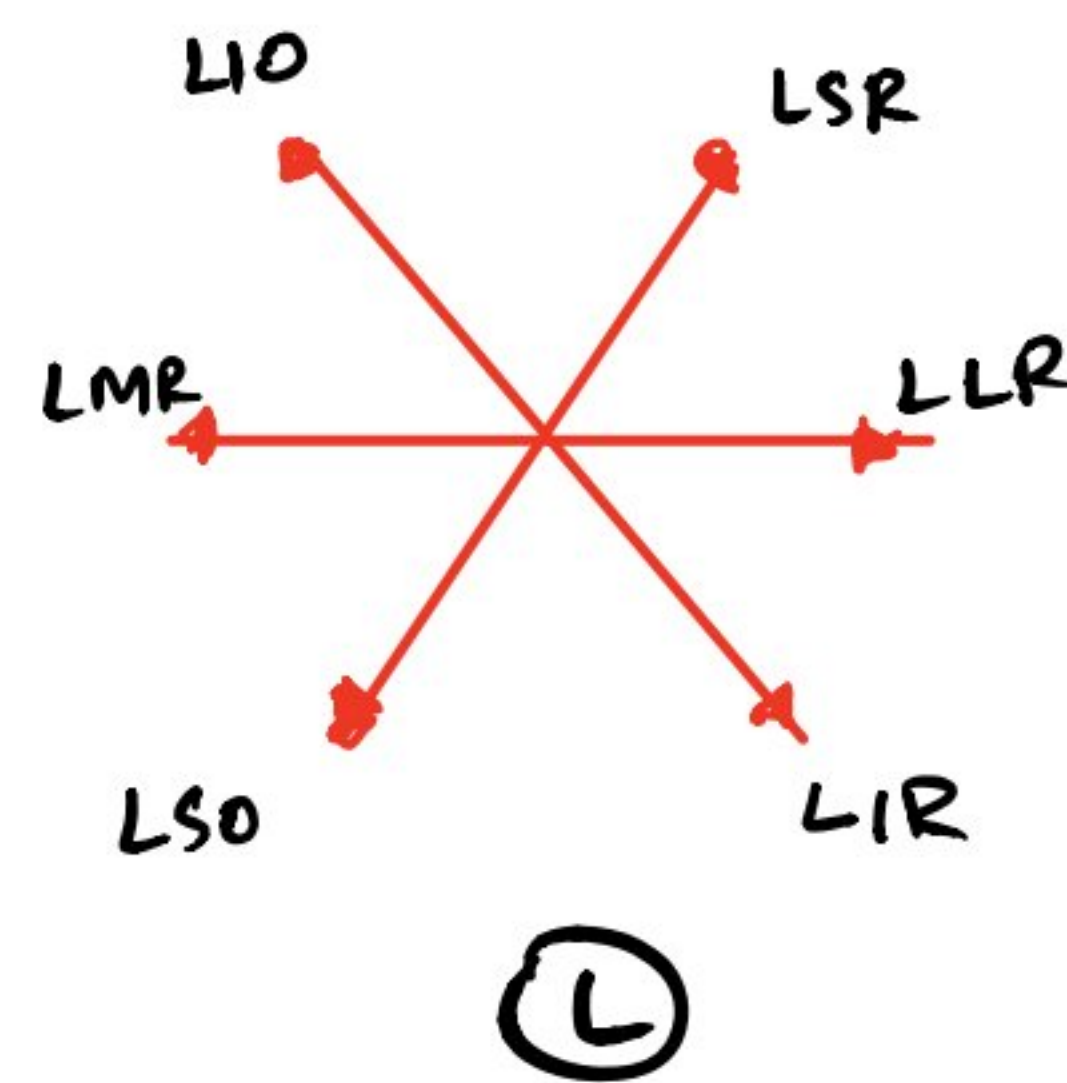
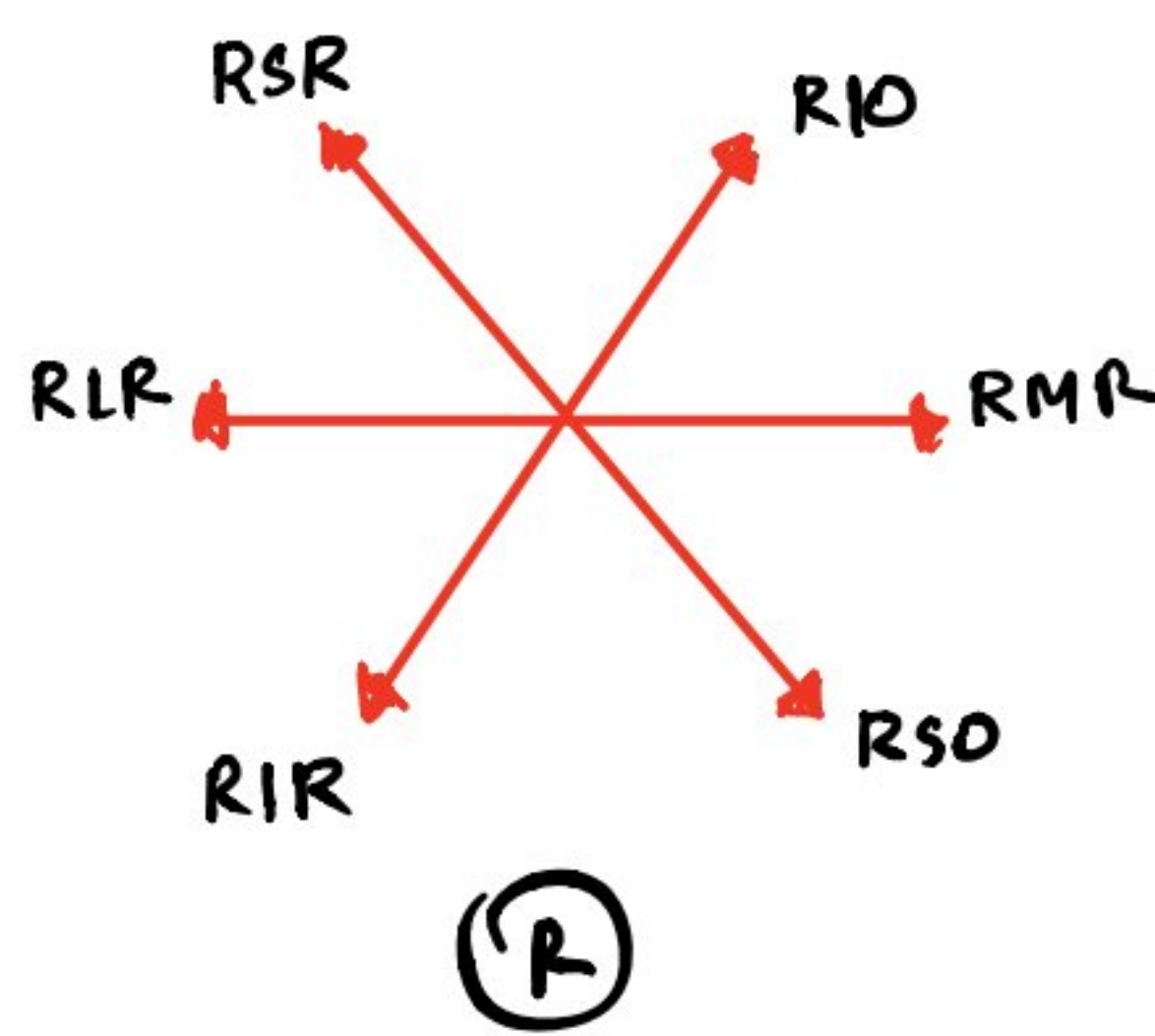
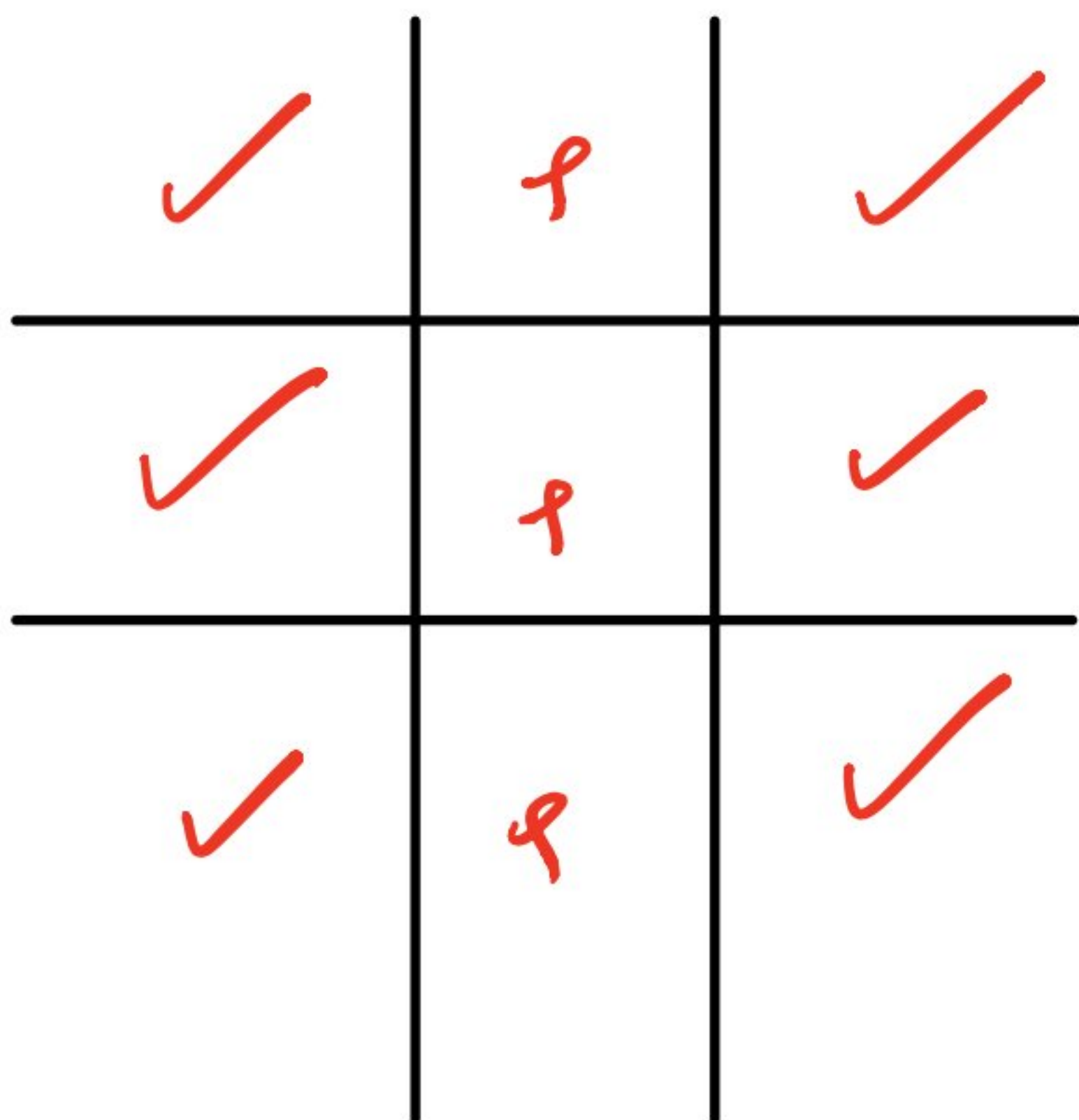
Hering's law of Equal innervation



for BSV → equal power req.
motor correspondence



CARDINAL GAZE: = 6



Comitant Squint: Angle of squint same all 9 gaze.

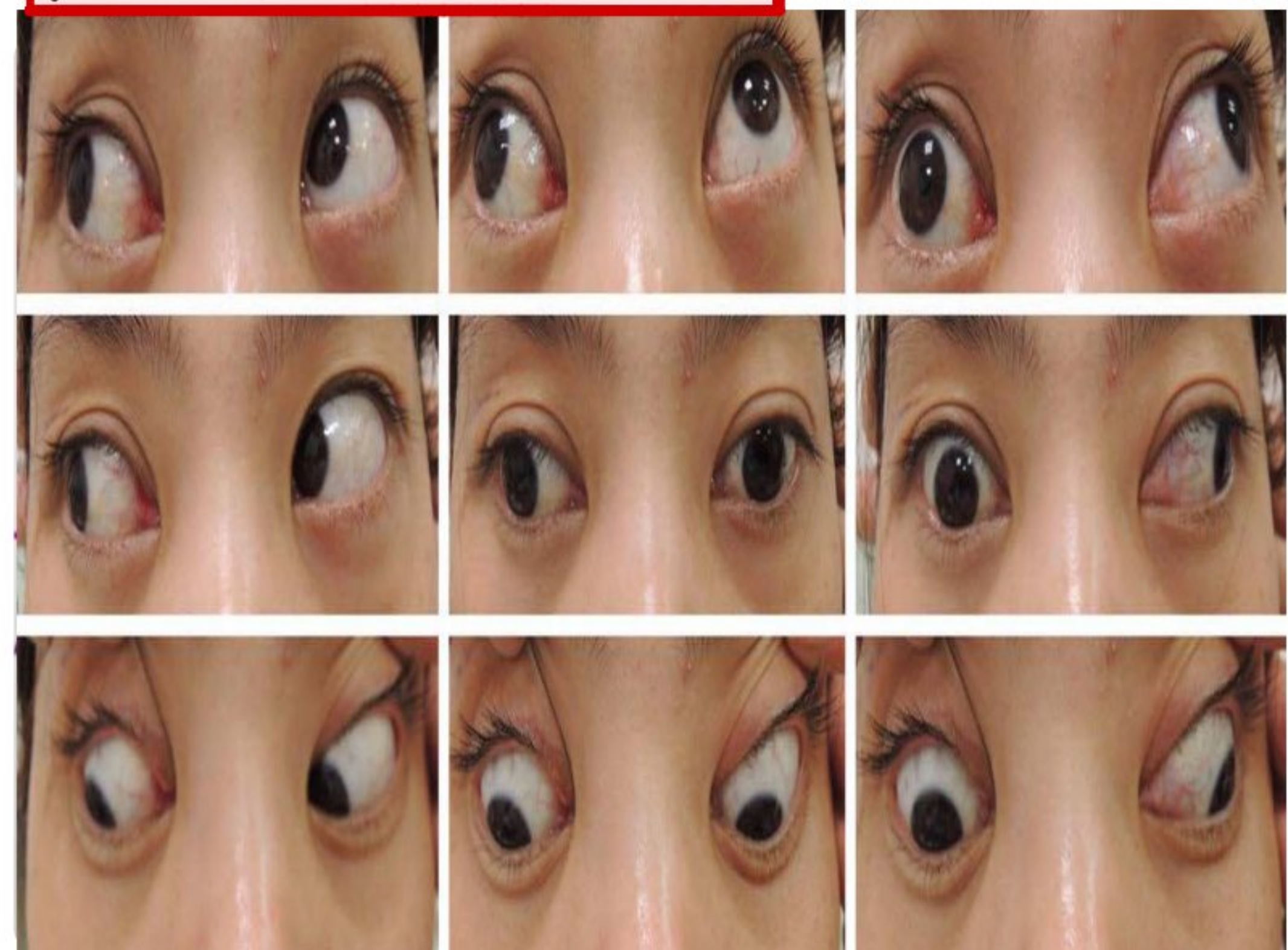
Incomitant Squint: " " different in different gaze

- Eg: 1) Paralytic Squint
2) Restrictive Squint
3) A-V pattern (squint) heterophoria



Fig. 1. Preoperative 9 cardinal photographs of a 3-year-old boy presenting with 50 PD of left esotropia without lateral incomitance secondary

INCOMITANT SQUINT

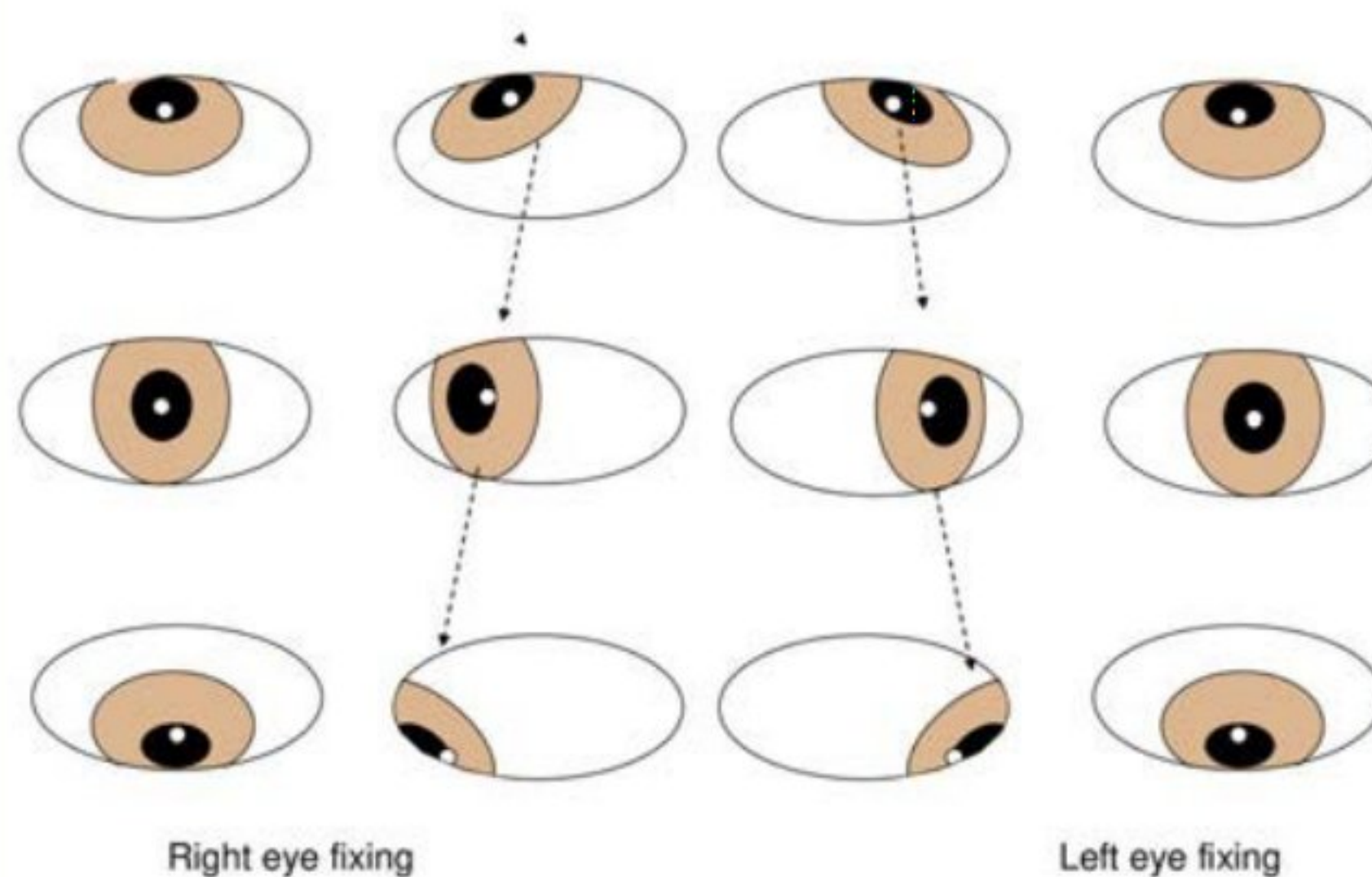


A Pattern Esotropia

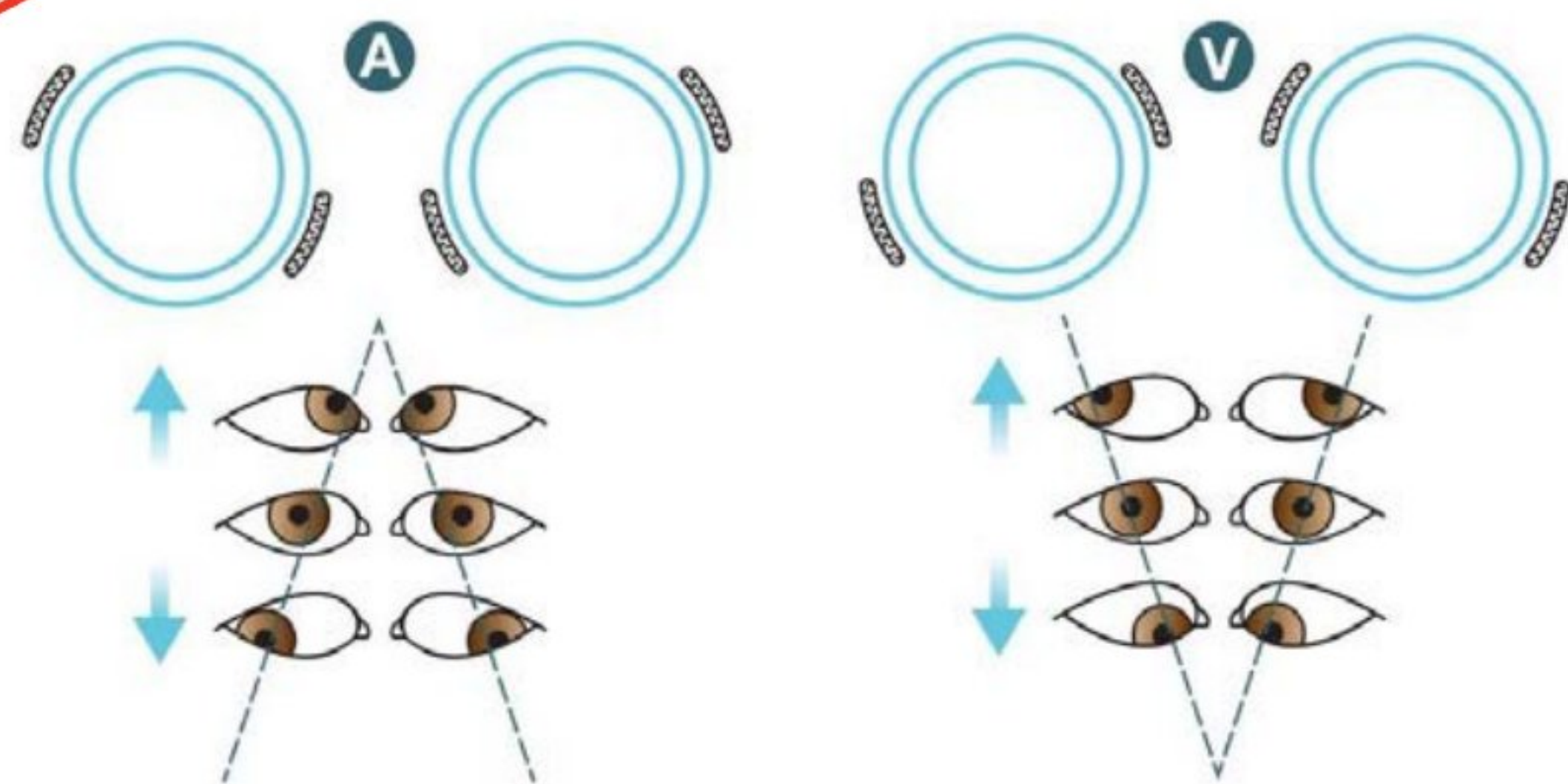


Dr Sudhir Singh

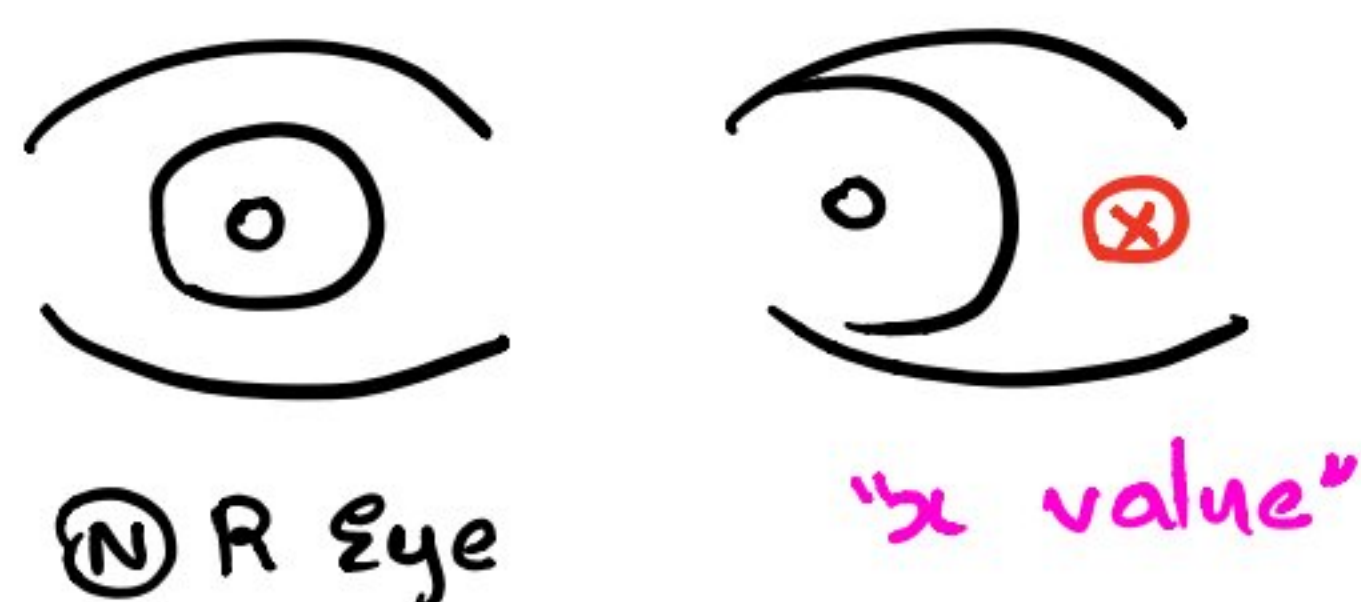
V - pattern esotropia



Q1 Q

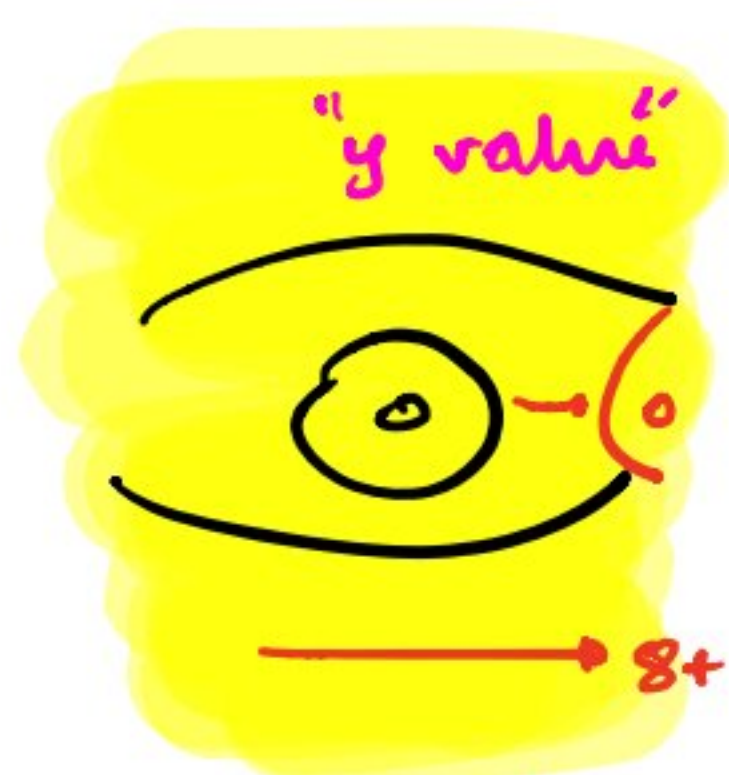


Primary deviation : Angle of squint when (N) eye is fixating
Eg: (L) LR Palsy

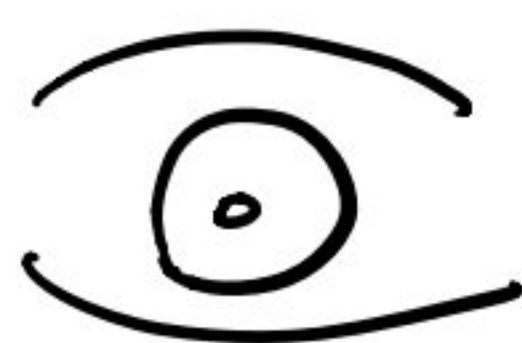


Secondary deviation : Angle of squint when paralytic eye is forced to fixate

Eg:



According to Hering's law
(N) (R) MR gets 8+



weak (L) LR [forced to fixate]

4+ → 8+
{ by brain }

In Paralytic Squint : $2^\circ \text{ deviation} > 1^\circ \text{ deviation}$
{ d/t to Hering's law of Equal innervation }

- Binocular diplopia
- Compensatory head posture { to avoid diplopia in 1st gaze }
- Incomitant / non comitant squint
- FDT -ve

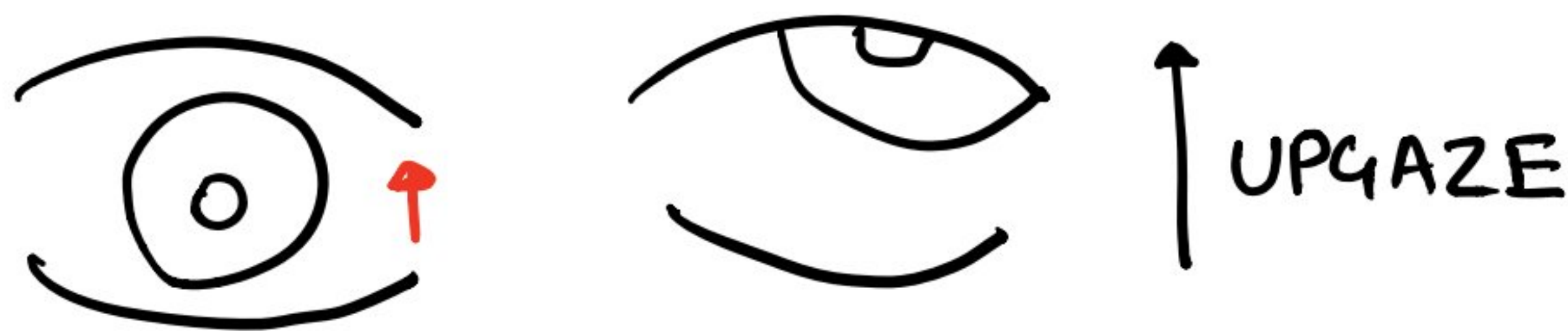
Q: Which among the following is (T) about concomitant squint?

- Primary deviation > 2° deviation
- 1° deviation < 2° deviation
- 1° deviation = 2° deviation
- None

sol. In a concomitant squint
Primary deviation is equal to secondary deviation

FORCED DUCION TEST

Ⓡ SR palsy ?
or
Ⓡ IR restriction ? } To find out
do FDT [Force duction test]

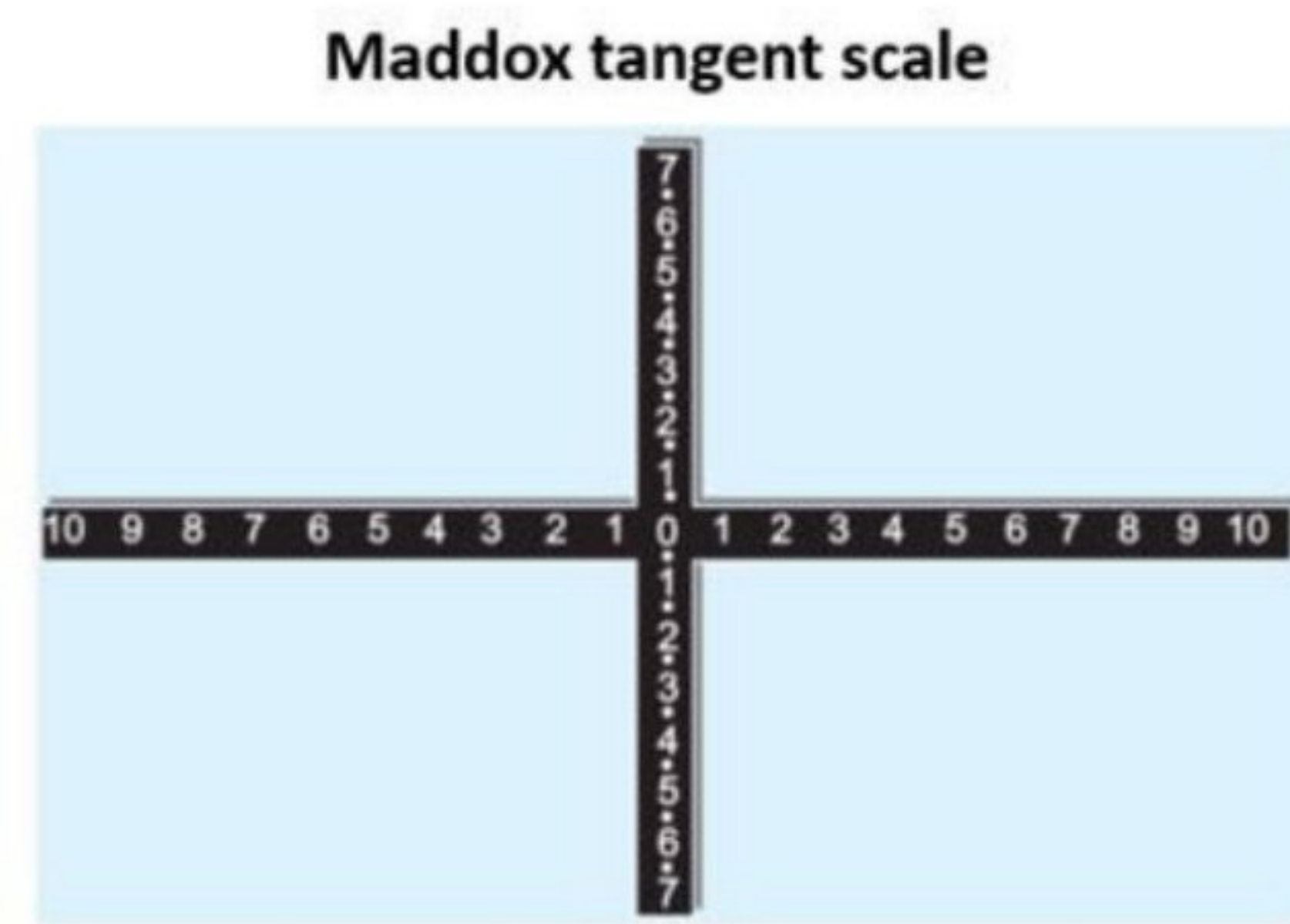


Poss. Problem : i, ↑ SR Palsy
ii, Problem which preventing eyeball from moving up
i.e IR muscle is holding the eyeball down.

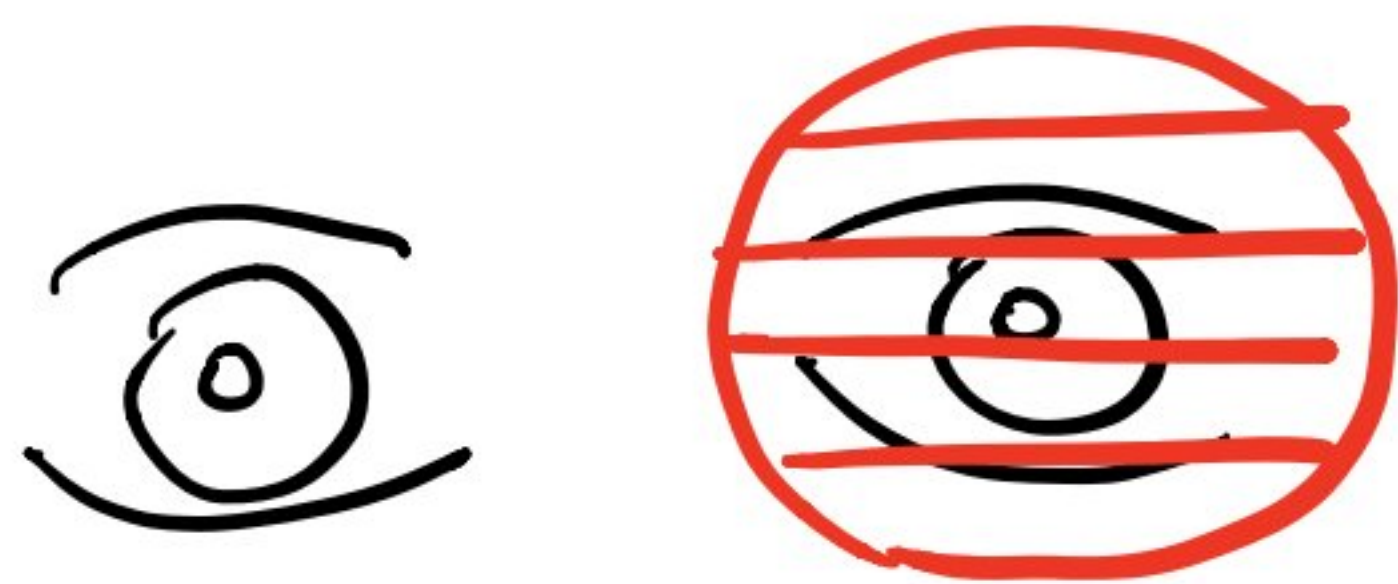
FDT ⇒ Ⓡ SR palsy ⇒ easily move the eyeball = FDT -ve
or

Ⓡ IR restriction ⇒ feeling of resistance / tug = FDT +ve

FDT
+ve Restrictive type of squint
-ve Paralytic squint



DIPLOPIA:



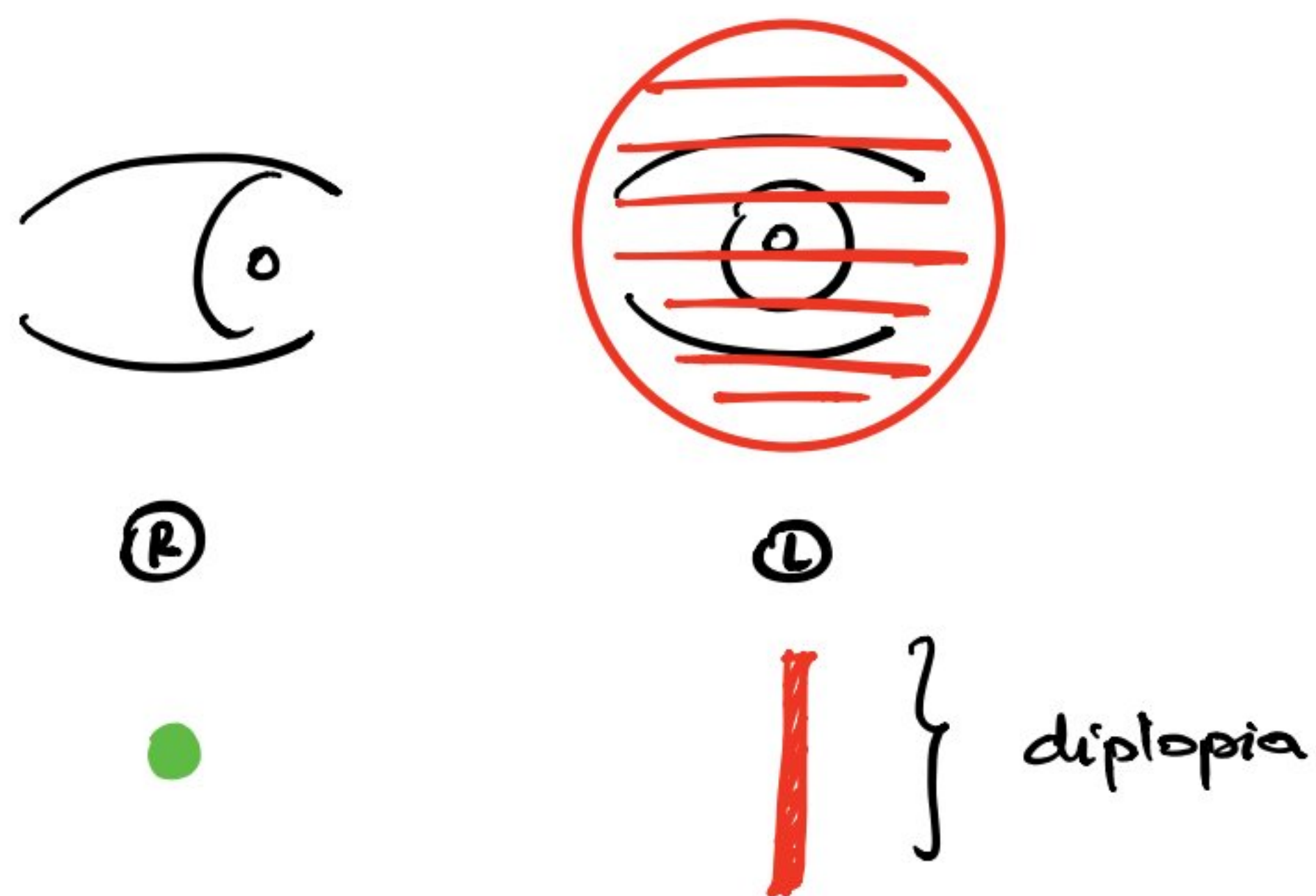
views vertical line



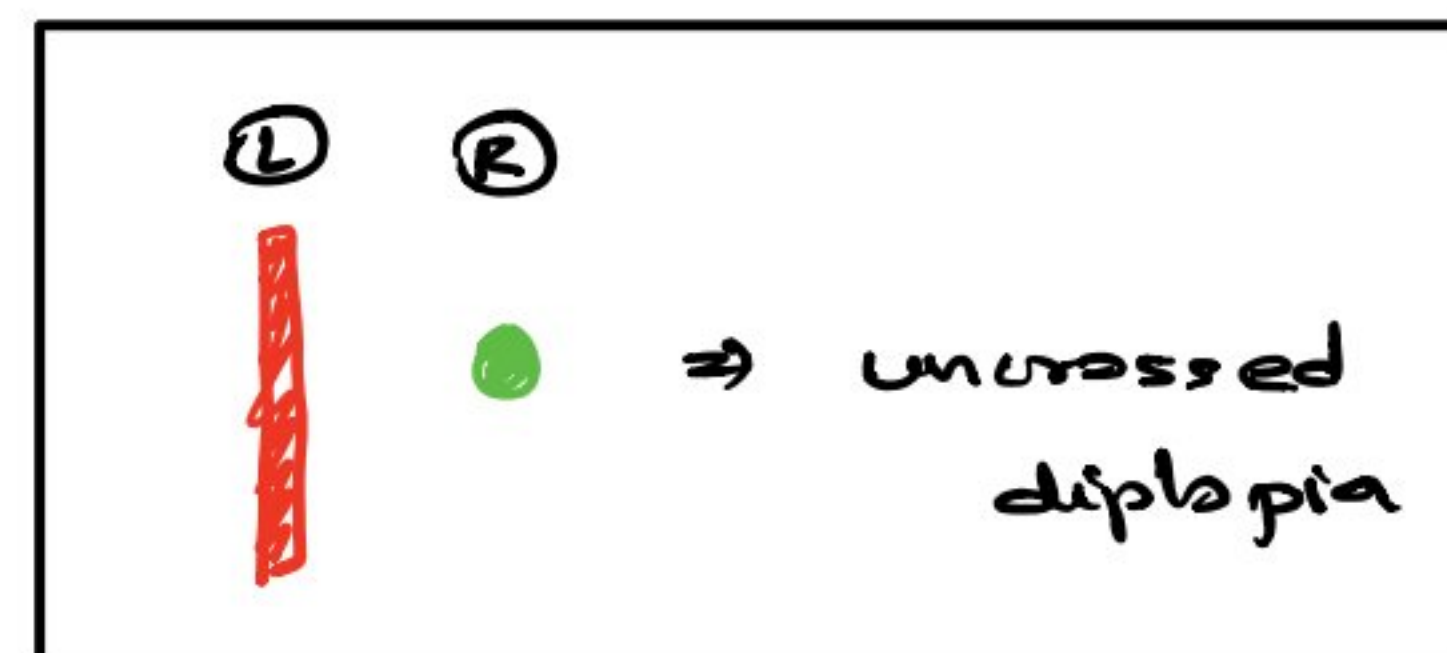
views horizontal line.

BSV = point source light

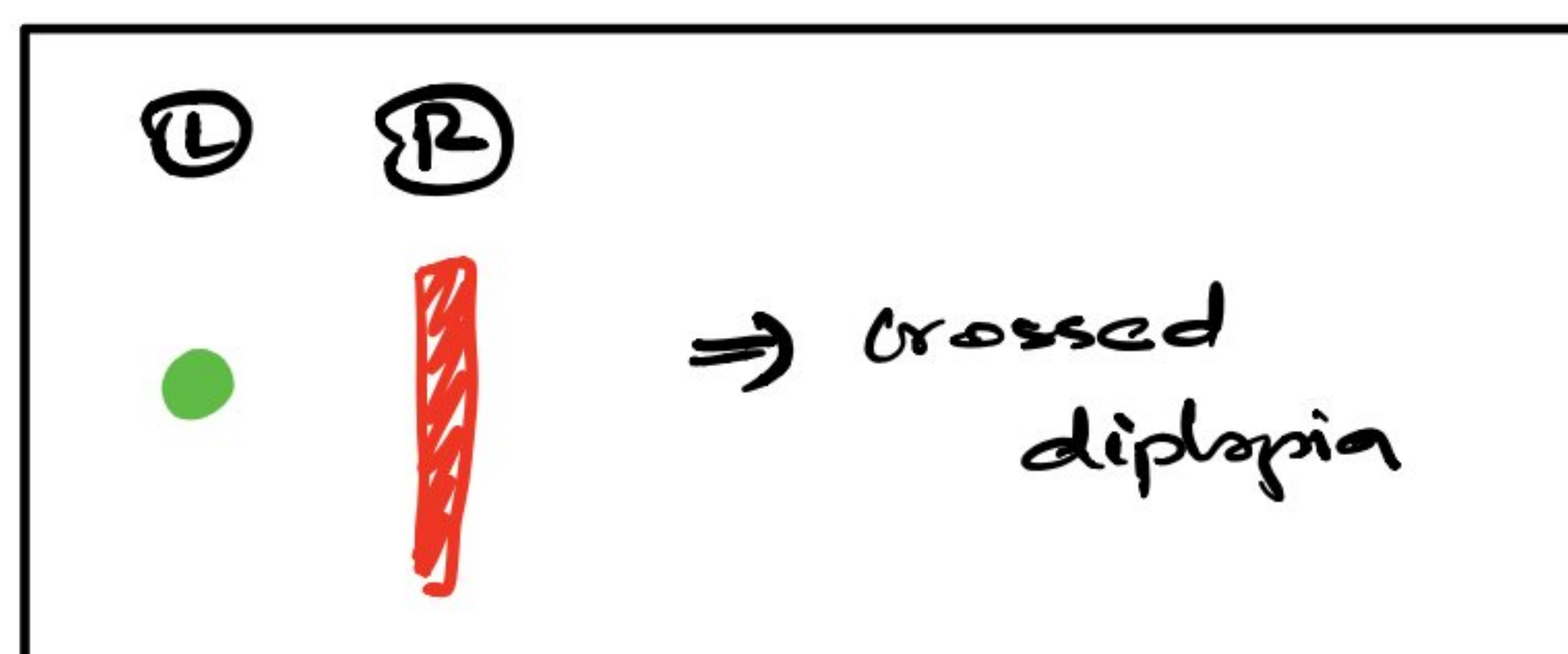
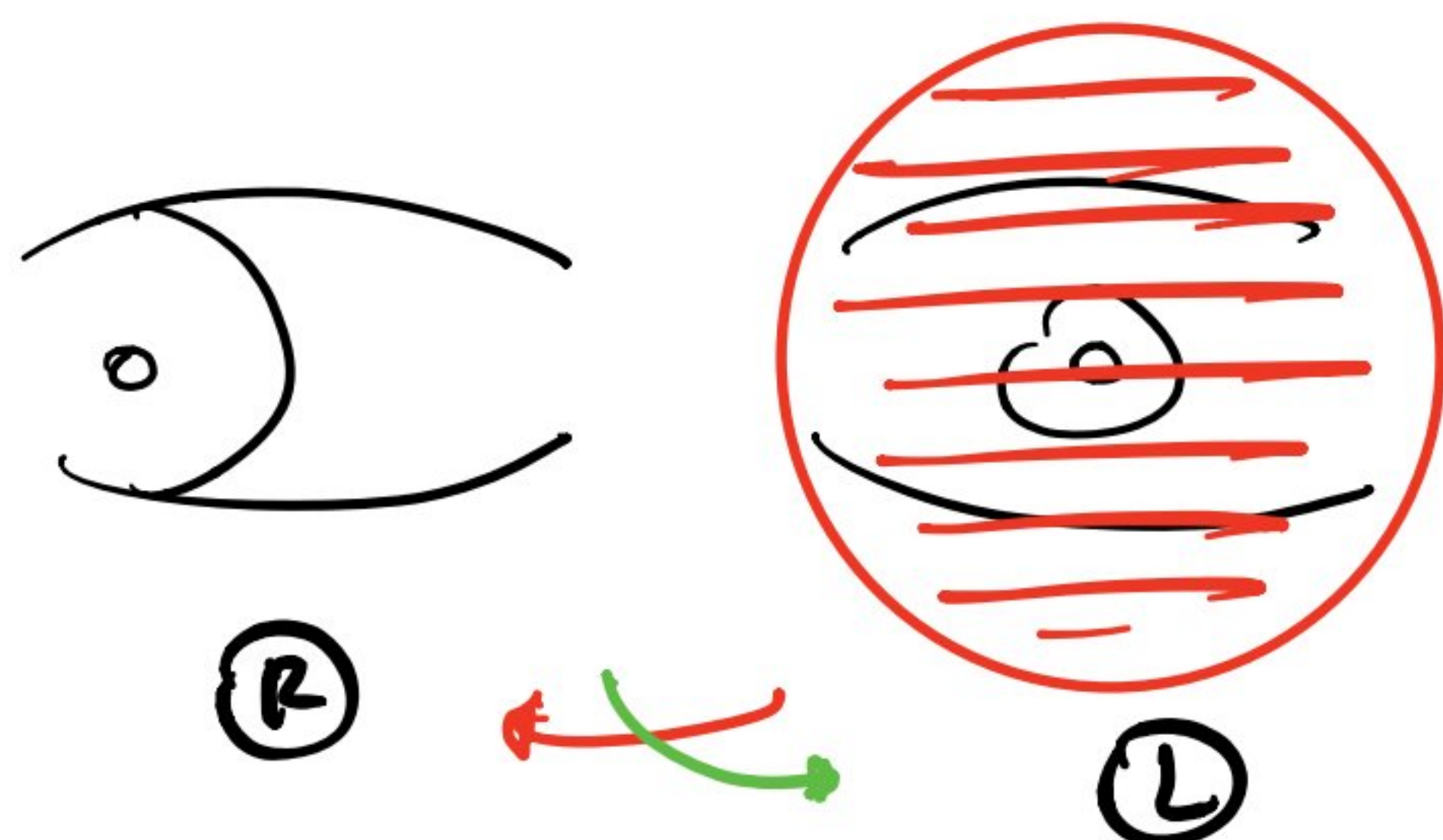
Convergent Squint : ESO



representation on paper



Divergent Squint : EXO

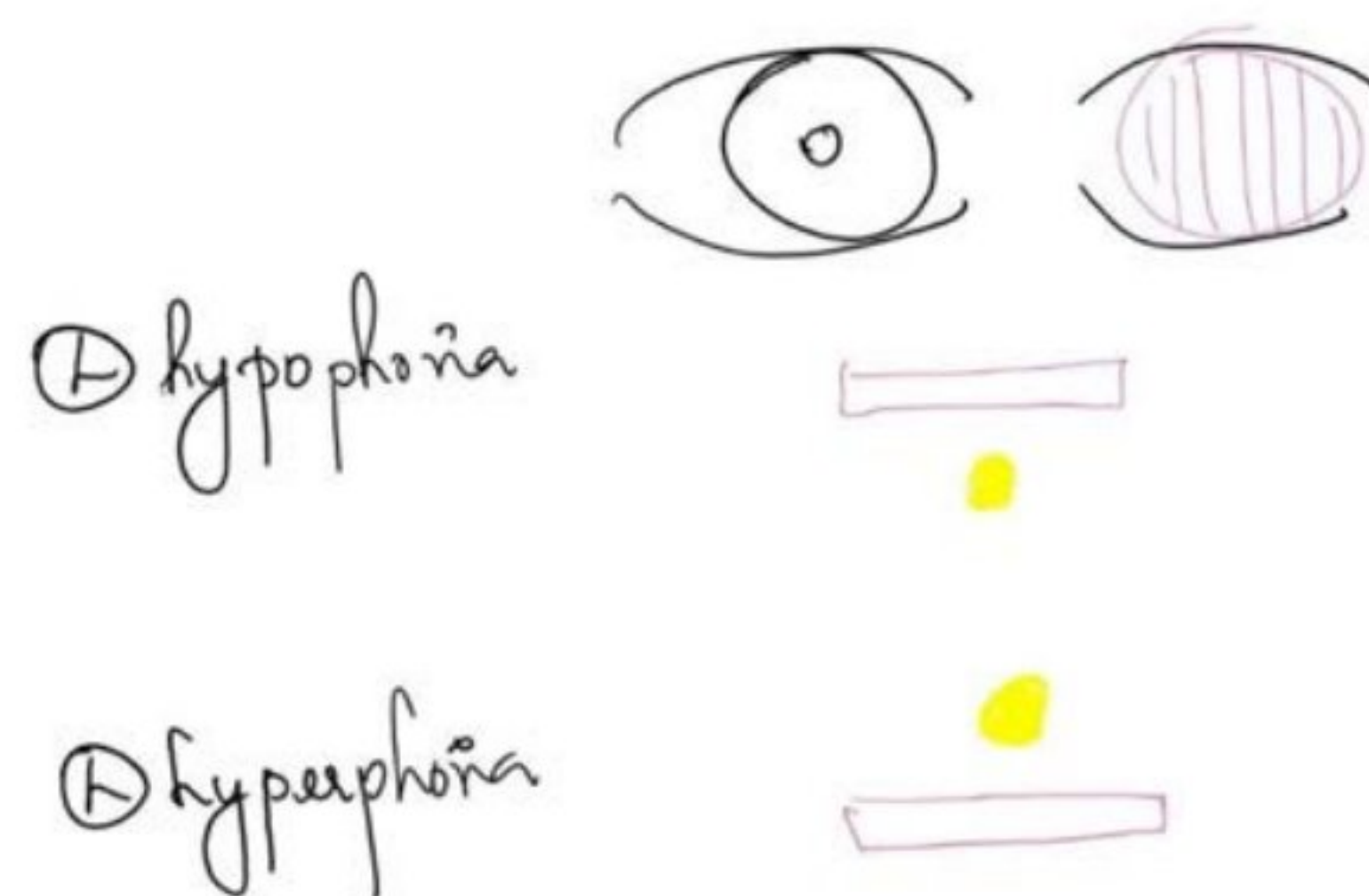


Management of Squint

- 1) Correction of Refractive error
Myopia $\rightarrow$ Exophoria (later, exotropia)
Hypermetropes $\rightarrow$ Esotropia

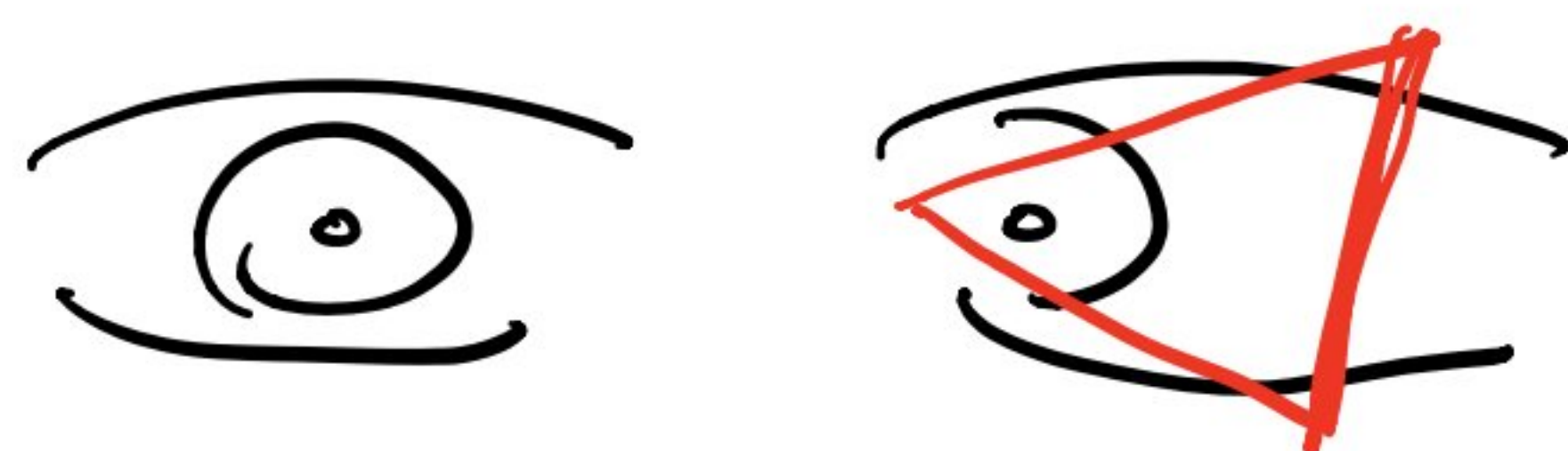
2) PRISMS $\rightarrow$ In glasses

$\hookrightarrow$ apex always towards deviation



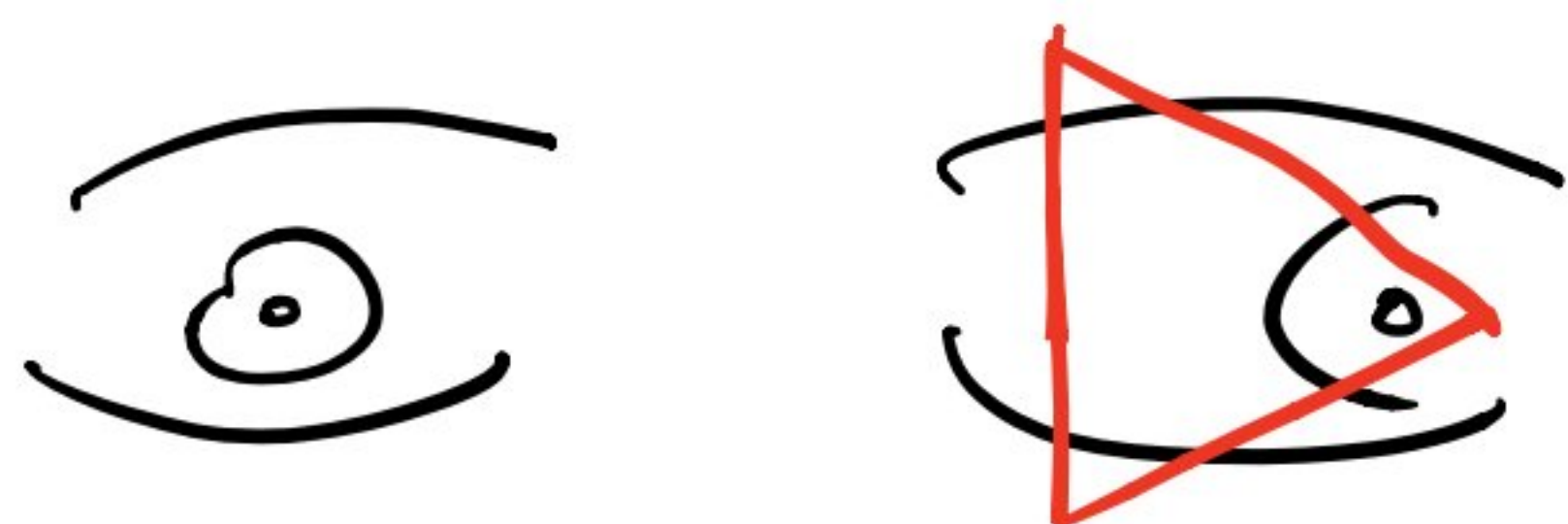
① Esotropia

Rx : ① 15 PD Base out prism



① exotropia

Rx : ① 15 PD Base in prism



③ Botulinum Toxin A : Injection into the overacting antagonist $\rightarrow$ correction of squint in 1st gaze

- b) Surgery : a) To strengthen the wk muscle → RESECTION
b) To weaken the strong overacting antagonist → RECESSION

Eg: ① LR Palsy

① LR Resection

±

① MR Recession

Q: Yoke muscle pair among the following is _____?

- a) Right superior oblique (SO) and left inferior rectus (IR)
 - b) ② superior oblique and ① infer. oblique.
 - c) ③ medial rectus (MR) and left medial rectus (MR)
 - d) ③ medial rectus (MR) and ④ lateral rectus (LR)
- sol. ③ SO and ② inferior rectus are the yoke muscle pair for levodepression.

Note:-
~

Q: Down beat nystagmus is associated with all except _____?

- a) Lithium toxicity
- b) Arnold - Chiari malformations
- c) Tumors of the third ventricle
- d) Cerebellar atrophy.

Q: Which of the following best defines "saccade"?

- a) Sudden jerky conjugate eye movements
- b) Smooth slow eye movmnt.
- c) Flicks
- d) Drifts

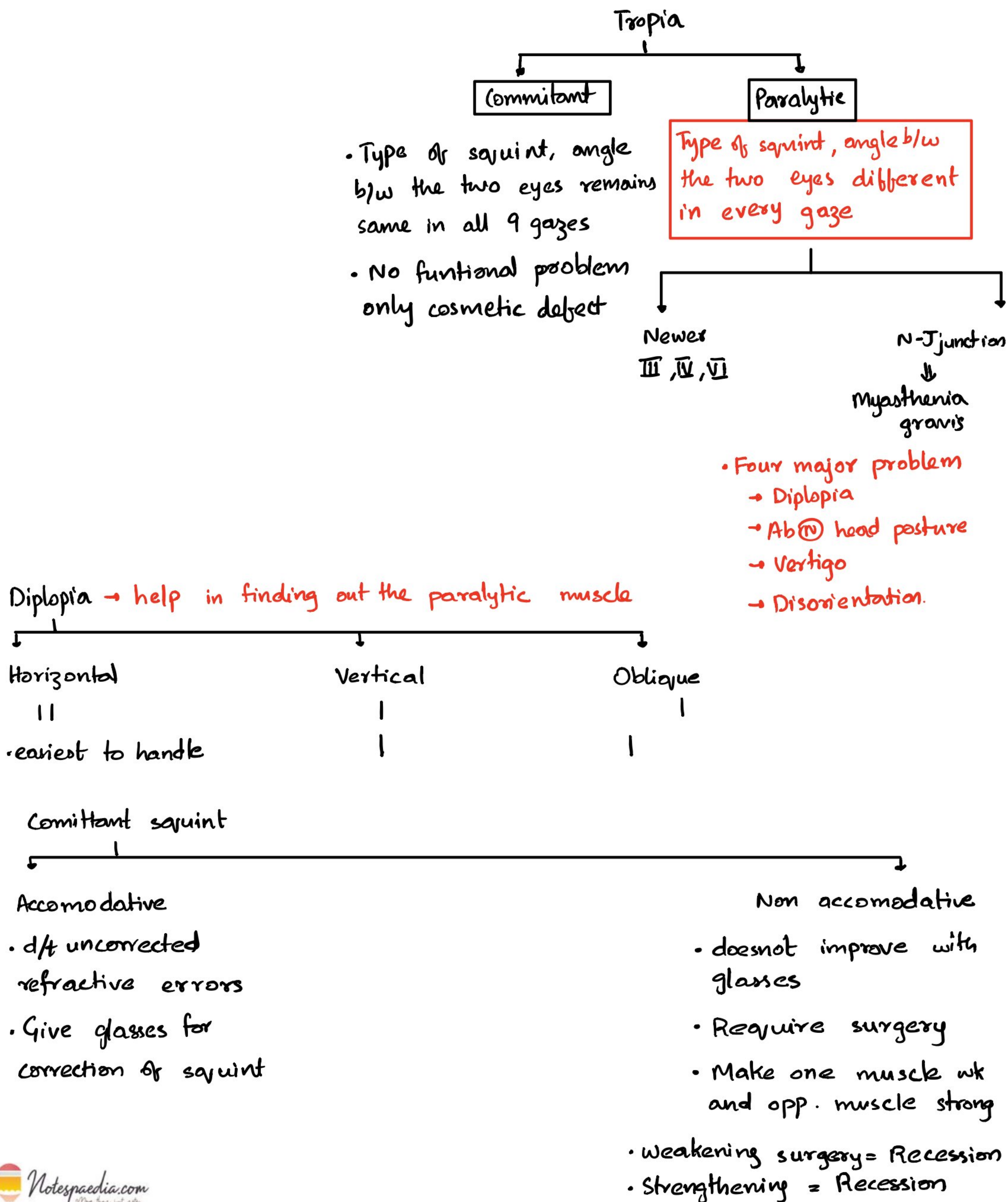
Note:-
~

Extras in squint

For phorias :: Required cover uncover test

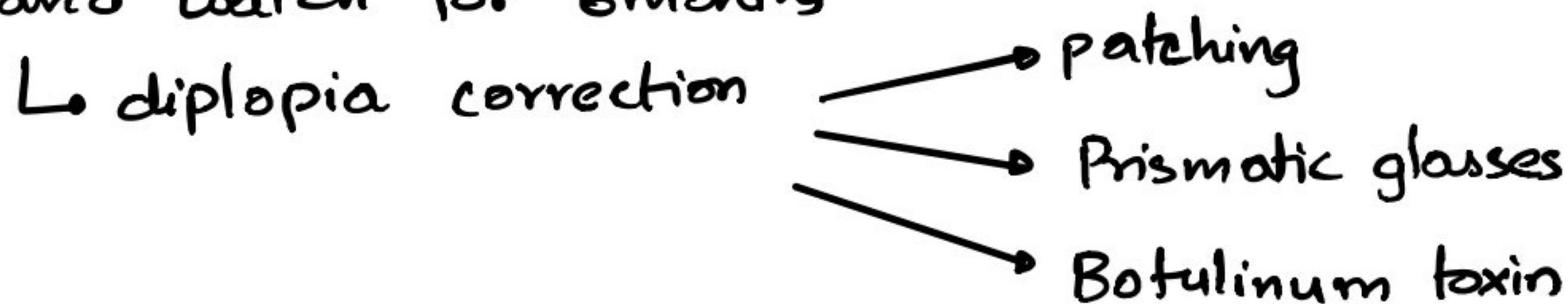
Tropia :: Require Rx

Phoria :: No Rx



Paralytic Squint Rx

- 1) Control the underlying disorder
- 2) Wait and watch for 6 months



Botax for wrinkles { "crow feet" → fine wrinkles on eye
• Root of nose wrinkles
• Forehead wrinkles

Inject botulinum toxin in opp. muscle of paralysis

Diagnosis:

3rd N. Palsy → Down and Out eye and ptosis

4th N. Palsy → Up and in eye & opp. side head tilt

4th C.N

- Longest intra-cranial nerve = 75mm
- Thinnest
- 1st one to get ruptured in closed head injury
- Only nerve to cross over
- Only CN to come out from dorsal surface of brain.

6th Nerve: Eye in, face out.

Myasthenia Gravis: • Old men, young girl

• ptosis

(1st muscle affected - LPS)

NYSTAGMUS: Repetitive involuntary oscillations of eye

3 type → Horizontal
→ Vertical
→ Torsional

- Jerk → fast component - side and slow drift
- Pendular → fast component - up/down [velocity equal in each direction]
- Mixed → pendular in 1^o gaze
→ jerk in lateral gaze

cause: → Pathological
→ Physiological

Physiological Nystagmus

- 1) Optokinetic : . induced OKN drum
 - slow smooth movement $\Rightarrow$ K/a pursuits
 - $\hookrightarrow$ controlled by 1/2 occipito parietal lobe
 - fast rapid movement in opposite direction $\Rightarrow$ K/a saccade
 - $\hookrightarrow$ controlled by 4/6 Frontal lobe

2) End gage Nystagmus

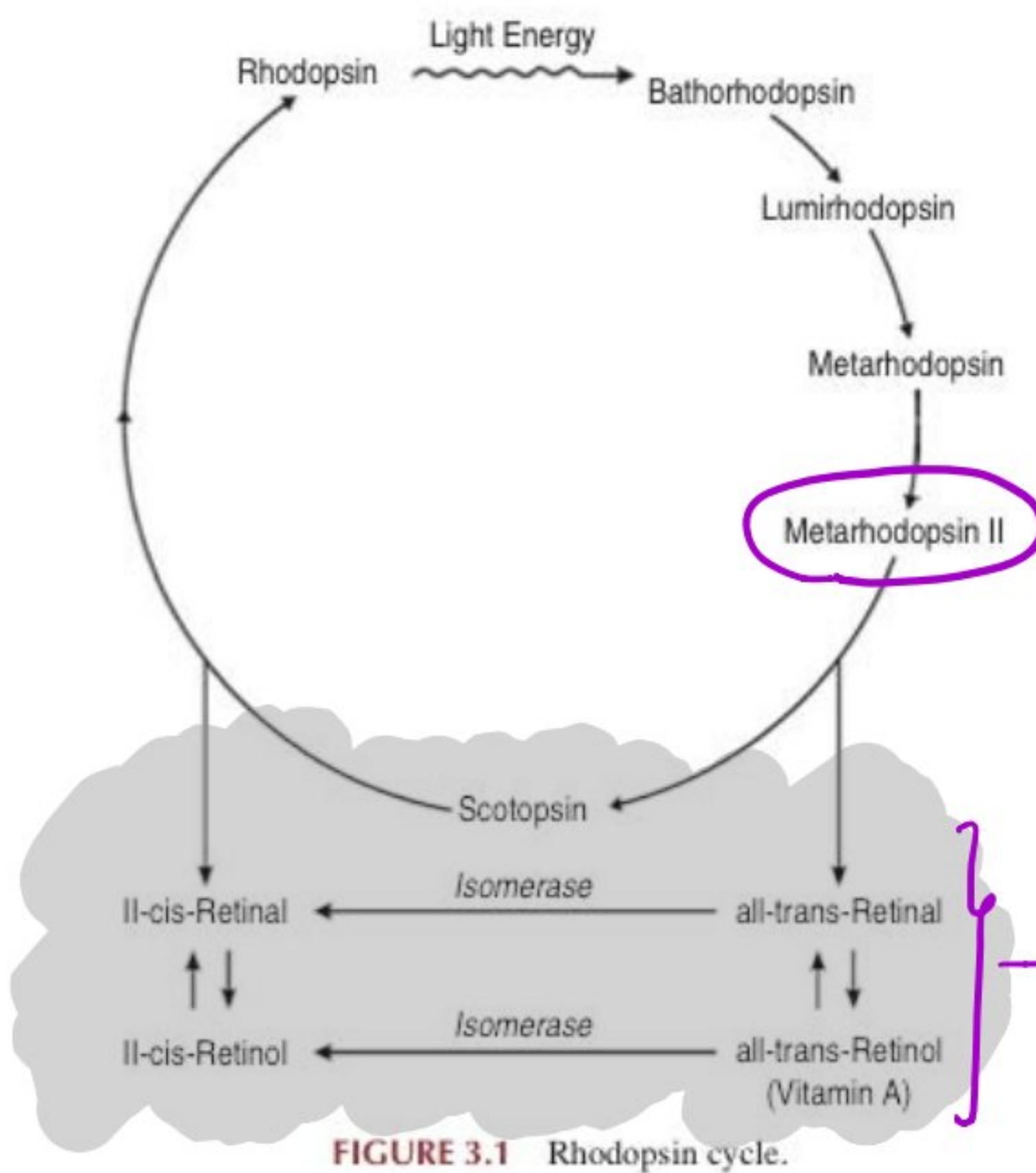
- 3) Vestibular Nystagmus [cows]
 - cold water in (R) $\rightarrow$ (L) Nystagmus

Pathological Nystagmus

- 1) INO $\rightarrow$ Ataxic Nystagmus
- 2) Optic chiasmatal lesion $\rightarrow$ see saw nystagmus.
- 3) Parinaud Syndrome $\rightarrow$ convergence, retraction nystagmus
- 4) Arnold Chiari's $\rightarrow$ Down beat nystagmus
- 5) Post. Fossa lesion $\rightarrow$ Up beat nystagmus
- 6) In Mineirs $\rightarrow$ Rotatory / pendular nystagmus
- 7) Infantile $\rightarrow$. $\downarrow$ on convergence
 - reversal Q OKN
 - $\hookrightarrow$ Exotropia $\rightarrow$ latent Nystagmus
- 8) Spasmus mutans $\rightarrow$ a/w Head nodding
- 9) Sensory nystagmus $\rightarrow$ d/t sensory deprivation

Rhodopsin Cycle:

- Visual pigment in the outer segment of rods responsible for scotopic vision.
- Role of Rhodopsin in the signal transduction cascade of vision is to activate **TRANSDUCIN**: G protein upon absorption of light



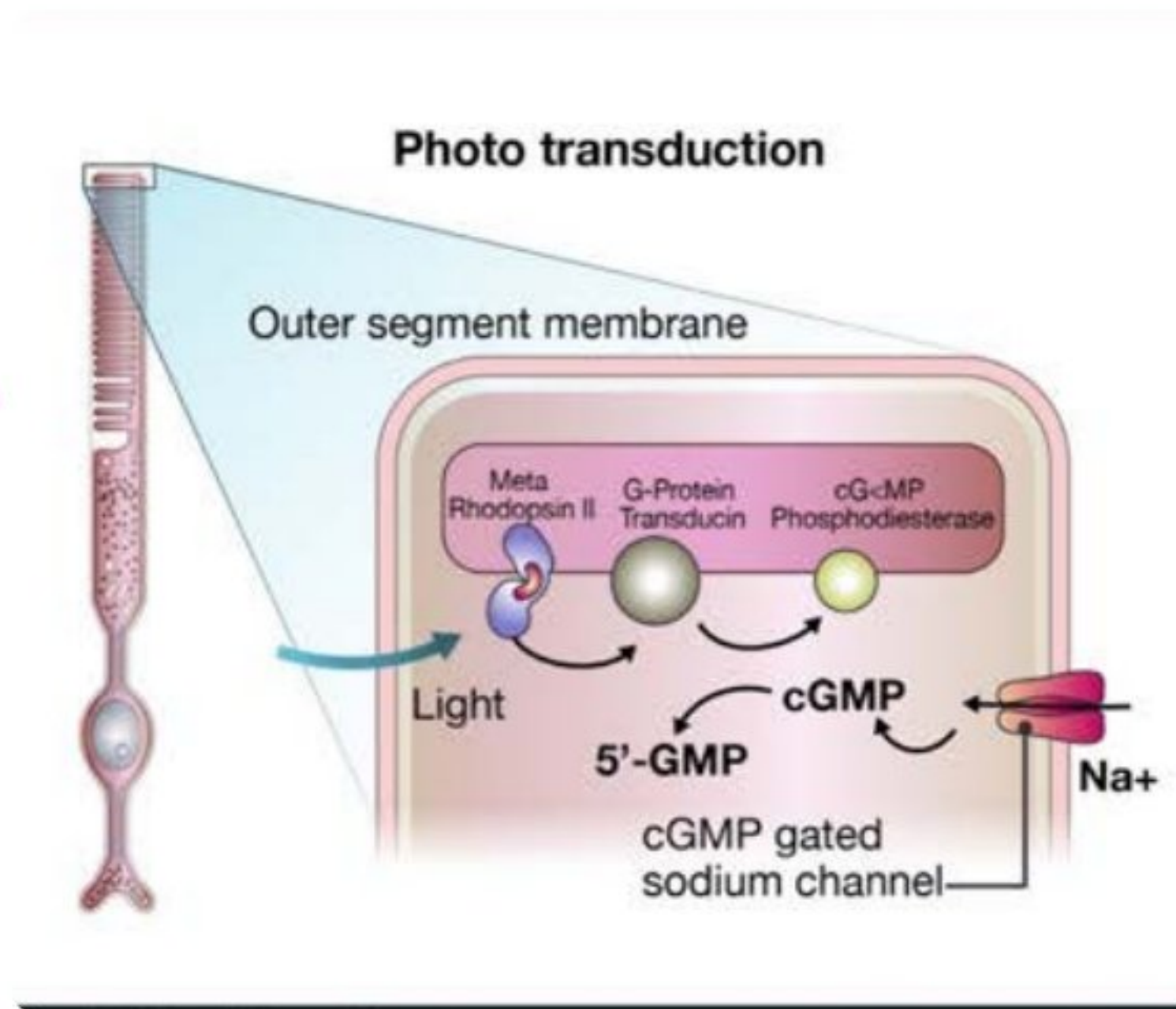
during dark
@ RPE layer

Metarhodopsin II activated Transducin
G protein $\rightarrow$ cGMP
phosphodiesterase

5'GMP $\leftarrow$ cGMP

In this process of converting cGMP to 5'GMP
the cGMP gated Na^+ channel closes

$\rightarrow$ This results in hyperpolarisation
or PHOTOTRANSDUCTION

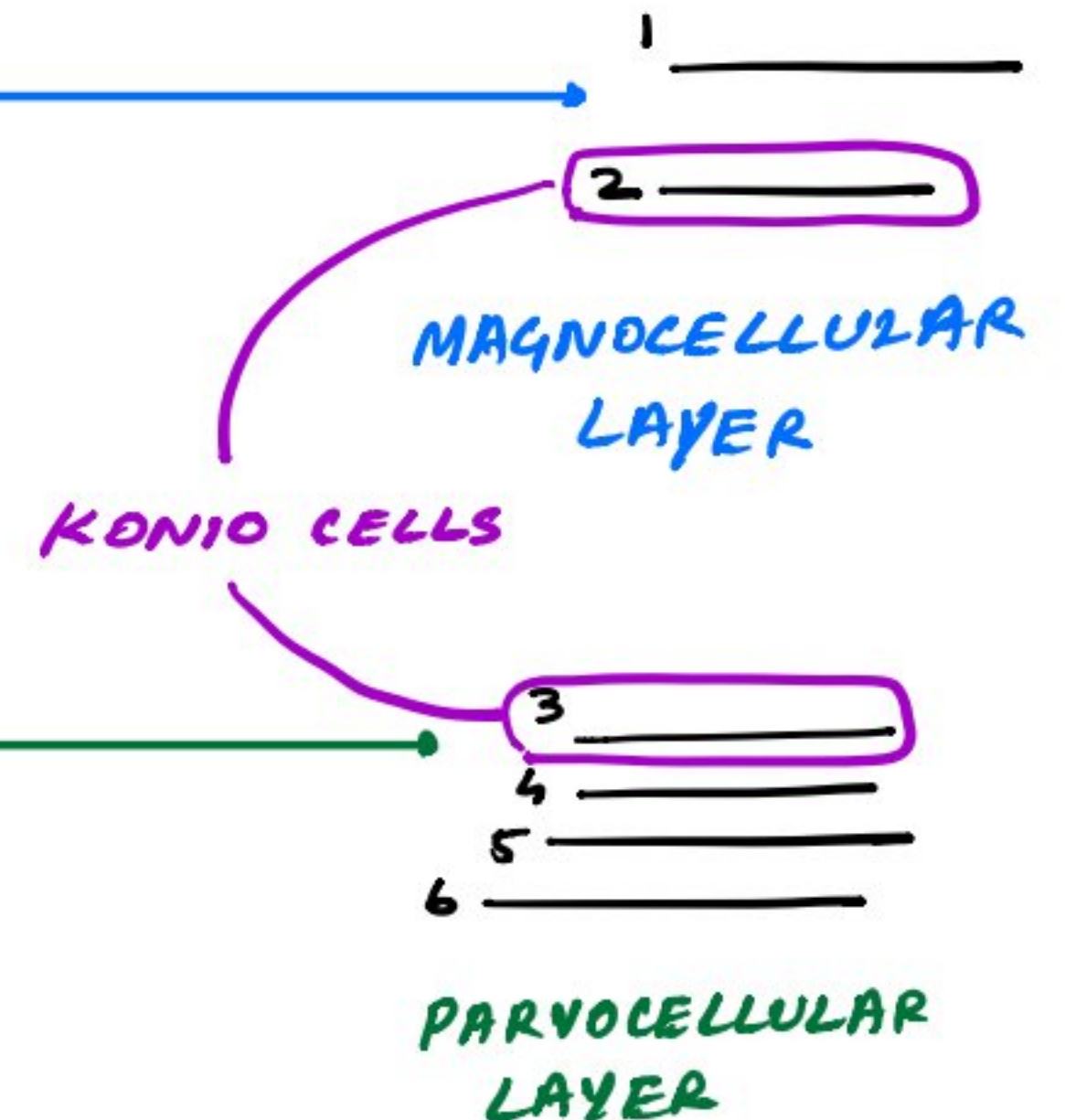


MAGNOCELLULAR, PARVOCELLULAR & KONIOCELLULAR PATHWAYS:

L4B

Retinal ganglion cells $\rightarrow$ M cell $\rightarrow$ movement
depth (stereopsis)
flicker
spatial organisation

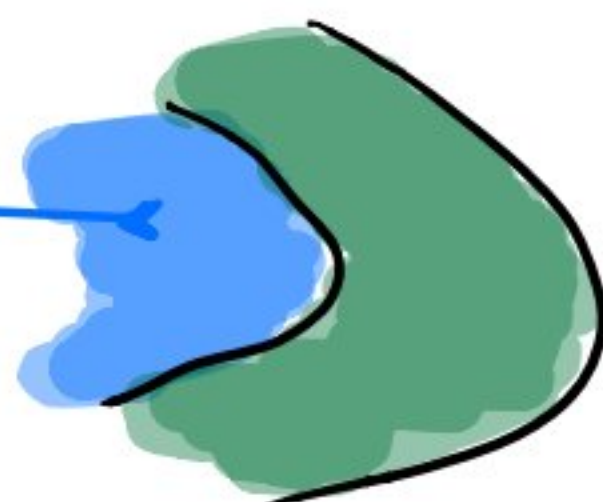
P cell: Macular predominance $\rightarrow$ color sensitive
90%
texture, shape
fine details.



L4B

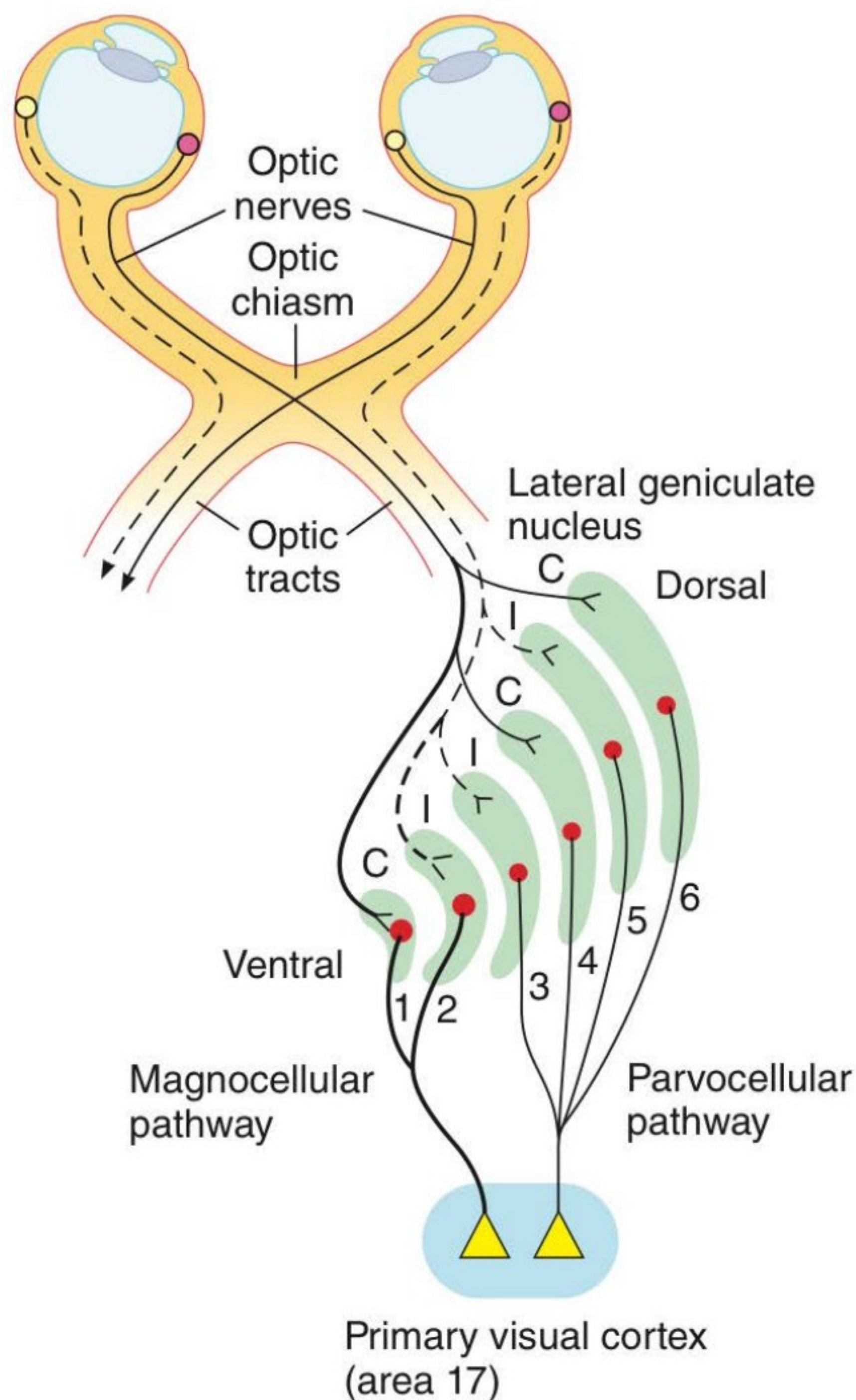
MAGNOCELLULAR
LAYER

VISUAL CORTX



Lateral geniculate	Striate cortex (V1)	Extrastriate cortex (V2)	Higher centres for body
Magno cells	→	→	Motion and stereo
Parvo cells	→	→	Form
Konio cells	→	→	Colour

FIGURE 3.2 The primary visual cortex, the striate cortex, transforms information received from the lateral geniculate nucleus and disperses this to appropriate areas in the extra striate or secondary visual cortex for transmission to higher visual areas where specific visual phenomenon is registered (as proposed by Livingstone and Hubel).





Check out our awesome notes

NEURONOTES

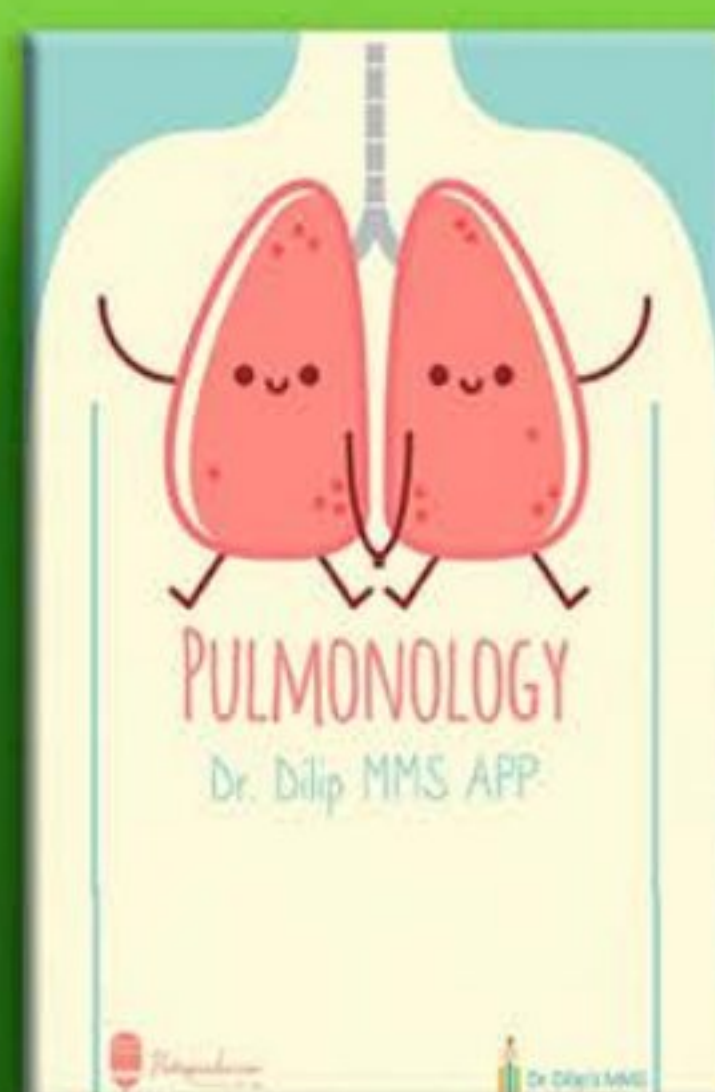


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